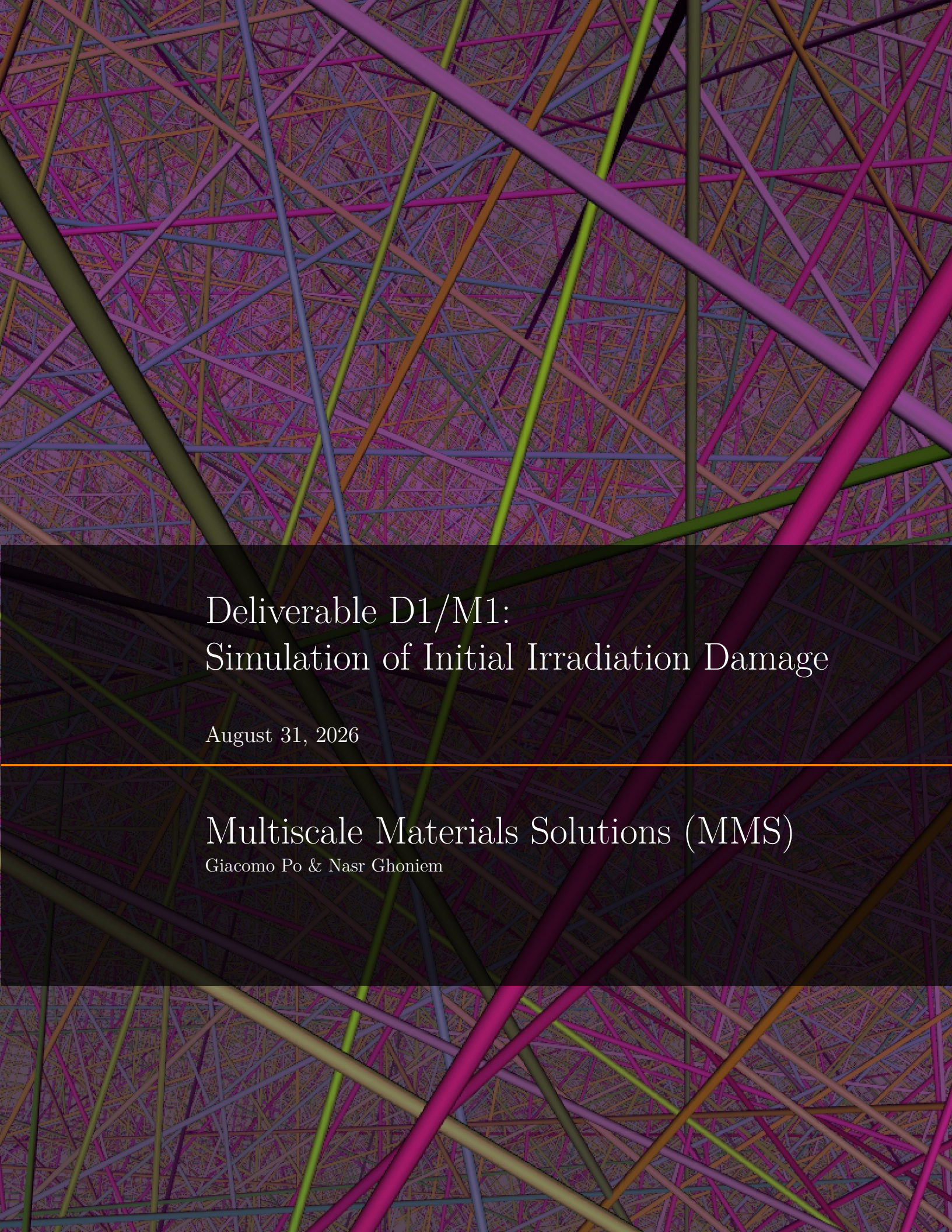




Deliverable D1/M1: Simulation of Initial Irradiation Damage

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Abstract

This report documents progress on Deliverable D1, *Multiscale Simulation Methodology*, and on Milestone D1/M1, *Simulation of Initial Irradiation Damage*, whose objective is a spatially resolved Cluster Dynamics–Discrete Dislocation Dynamics framework for irradiation-induced defects in zirconium. Two codes implement one model. **ZrMicro-0d** integrates nineteen coupled rate equations for the mobile point defects, four dislocation-loop families and an evolving network density; **ZrMicro-3d** solves the same reaction network on a finite-element mesh, treating the mobile species as a quasi-steady reaction–diffusion problem and marching the immobile populations pointwise. The spatial code began from a baseline with no immobile population; loop content, loop density and the kinetics that evolve them were added in turn. A parameter identification needs thousands of forward solves, so the reduced model serves as a cheap surrogate: twenty-three parameters are identified against a temperature- and dose-resolved database of loop densities and diameters, by a normalized group-wise objective minimized with a hybrid global, surrogate and local search, and the result is transferred to the three-dimensional material file.

The calibrated reduced model reproduces the build-up, overshoot and saturation of the defect kinetics, the prismatic $\langle a \rangle$ densities and their thermal suppression, and the saturated $\langle a \rangle$ diameter at low to intermediate temperature. The principal quantitative gap is the basal $\langle c \rangle$ diameter, which saturates below the measured value although its number density is matched, pointing to missing physics in vacancy-loop coarsening. The transfer of that calibration is demonstrated rather than assumed: **ZrMicro-3d** reproduces the **ZrMicro-0d** rate coefficients to within 0.1% and the interior mobile concentrations and $\langle c \rangle$ density to within a few percent, every verification ratio staying constant in dose from about 5 dpa — the signature of a structural difference rather than a unit or rate-constant error, which the loops would integrate into a drift. The spatial predictions, interpreted only then, resolve what the mean field averages away: a denuded zone at the absorbing grain boundary that thickens with dose, many small loops near the boundary against fewer large ones in the interior, and stored content correspondingly *enhanced* at the boundary for the prismatic families and *depleted* there for the basal one, which carries into grain-size-dependent irradiation growth. The residual code-to-code differences are diagnosed rather than resolved: the $\langle a \rangle$ density in **ZrMicro-3d** sits at about half the reduced-model value, consistent in sign with a homogeneous clustering flux not yet wired into loop nucleation, and, grain boundaries not yet being loop sinks, the near-boundary $\langle a \rangle$ density rises without bound and is not quantitative. The calculations are further restricted applied stress, isotropic diffusion and a bias fitted at 573 K, so the tensorial anisotropy mechanism is not yet exercised. Cross-verification itself proved diagnostic, exposing a Burgers-vector error and an uninitialized binding-energy read upstream.

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1 Introduction

The first deliverable under the current SOW is Deliverable D1: *Multiscale Simulation Methodology*. The objective is to develop and/or extend a Spatially Resolved Cluster Dynamics (SRCD)–Discrete Dislocation Dynamics (DDD) framework to simulate the nucleation and growth of irradiation-induced defect species in Zirconium. The framework shall generate continuous spatial fields describing defect species concentrations as functions of temperature and irradiation dose. Model parameters shall be calibrated to capture representative defect species and behaviors observed in Zirconium and its alloys. The methodology combines CD simulations to establish the irradiation-induced initial microstructure with a nucleation criterion for $\langle a \rangle$ dislocation loops once a threshold damage level (dpa) is reached.

In milestone D1/M1: *Simulation of Initial Irradiation Damage*, spatially resolved CD simulations shall be used to predict microstructural evolution, including defect nucleation and growth. Outputs shall consist of continuous spatial fields of defect cluster concentrations at prescribed temperatures and irradiation doses. Results shall be validated against experimental data across a range of irradiation conditions. We report here on our progress in the implementation of deliverable D1, specifically milestone 1 (D1/M1). The spatially resolved code used for this milestone did not arrive in its present form, and because much of what follows is only intelligible against the sequence of changes that produced it, that sequence is set out first, in Section 2.1.

2 Spatially Resolved Cluster Dynamics

MoDELib2-NNL implements a SRCD model of the irradiation defect microstructure, where defect clusters are represented by continuum fields. The method is called “**reduced**” because it resolves the concentration of a limited number of **mobile** defect clusters, each having a fixed defect content. Clusters of larger sizes, which are considered to be **immobile**, have their spatial description limited to their number density and total defect content, without resolving their size distribution explicitly.

2.1 Recent developments in the MoDELib2-NNL code

The spatially resolved model set out in the rest of this section was not present in the code at the start of this work, and the order in which it was built is worth following, because each stage is a self-contained capability and the limitations of the earlier ones explain the shape of the later.

The starting point is MoDELib2-NNL as distributed, which solves the steady-state reaction–diffusion problem for the mobile point defects and small interstitial clusters on a finite-element mesh, and nothing else. It carries no immobile population whatever: the immobile state is declared with `iSize = 0`, `solveImmobileClusters()` is an empty function body, there is no immobile integrator of any kind, and the source carries a literal `// Missing immobile sinks` placeholder where the loop contribution to the mobile equation would go. The dislocation loops are absent rather than simplified, so the mobile species see only the network and the domain boundary, and the microstructure cannot evolve at all.

The first stage opened the immobile state to four fields, `iSize = 4`, carrying one *content* equation per loop family — the stored defect content c_k — while the loop number density N_k was held fixed at its initial value. That is the smallest addition which lets the loops act back on the mobile species, because the sink strength $S_k = 2\pi r_k N_k$ needs a radius and the radius follows from the content through $r_k \propto \sqrt{c_k/N_k}$: with a frozen density, a growing content is a growing loop, and a growing loop is a stronger sink. It is enough for a spatially resolved growth calculation, in which existing loops absorb the biased defect flux and swell, and it is not enough for a microstructure. With N_k frozen there is no nucleation, no coalescence and no dissolution, so the population cannot respond to dose, to temperature or to the local defect fluxes, and the loop density must be supplied rather than predicted.

The second stage removed that restriction by promoting the density to a state variable of its own, doubling the immobile state to `iSize = 8` — four families, each carrying a number density and a content. A free density is only meaningful alongside the kinetics that change it, and these were added with it: cascade nucleation of loops at a finite birth size, geometric coalescence through the like-loop and loop–network Avrami channels, and thermal dissolution of the sub-critical vacancy embryos that fail to reach a stable size. Three further pieces of physics entered at the same time. The diffusional-anisotropy-difference capture efficiencies bias absorption by loop habit, so that prismatic and basal loops see different arriving fluxes from the same mobile field. The loop-absorption sink was made

mass-conserving, debiting the mobile species by exactly what the loop content gains, which is what allows the loop and point-defect balances to be closed against one another. And the network dislocation density became a state variable in its own right, fed by the line length of loops that coalesce into it, while the immobile Burgers vectors were assigned crystallographically so that the three prismatic variants and the basal family are distinguished by habit rather than by label.

That is the version described in the remainder of this section and used for every result reported here. Section 2.2 gives the mobile species and the steady reaction–diffusion equation they satisfy; Section 2.3 gives the bias factors and sink strengths through which the immobile populations act back on them; and Section 2.4 gives the immobile families themselves, the two moments each carries, and the equations that evolve them.

One point of provenance is worth recording, because the naming invites the opposite conclusion. A separate cluster-dynamics branch of the same code, `MoDELib-fullCD`, also carries eight immobile fields, together with a continuum-to-discrete loop conversion that the present branch does not have. It is a sibling of this work rather than its ancestor: it discretizes the immobile rates by a Galerkin projection at quadrature points where the present branch uses nodal collocation, and advances them by explicit Euler with sequential implicit loss factors where the present branch uses one fused semi-implicit update with sub-cycling. Parameter names are shared between the two because both descend from the same formulation, not because either was derived from the other.

2.2 Mobile defect clusters

The mobile defect clusters are defined by the `mobileSpeciesVector`, a vector parameter set in the material file. For example, `Zr_CD4.txt` defines `$ mobileSpeciesVector=-1 1 2 3`; so that four mobile species are considered: single vacancies, self-interstitials, di-interstitials, and tri-interstitials. We denote this vector $\mathbf{m}_M$; its entries are the *signed* cluster sizes m^k , with the polarity $p^k = m^k/|m^k| \in \{+1, -1\}$ selecting interstitial ($p = +1$) or vacancy ($p = -1$) character, and $|m^k|$ the number of point defects in the cluster. For the example above $\mathbf{m}_M = [-1, +1, +2, +3]^T$ and $\mathbf{p}_M = [-1, +1, +1, +1]^T$. As in Section 2.4, we use the boolean selectors $\mathcal{M}_i = \frac{1}{2}(1 + \mathbf{p}_M)$ and $\mathcal{M}_v = \frac{1}{2}(1 - \mathbf{p}_M)$.

MoDELib2 solves the following steady-state reaction-diffusion partial differential equation (PDE) for the mobile content field $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$, expressed in fractional (per-atom) units:

$$\mathbf{0} = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}. \quad (1)$$

Throughout this work, we distinguish two kinds of non-scalar quantities by their typeface. Upright sans-serif bold denotes arrays in *species space* (indexed by the mobile defect species $v, i, 2i, 3i$ or by loop family), with lowercase symbols for vectors, e.g. the concentration array $\mathbf{c}_M$, the stacked flux $\mathbf{J}$, and the vacancy-loop dissolution source $\mathbf{S}_{vL}$ (Eq. 53), and uppercase symbols for matrices, e.g. the reaction matrix $\mathbf{D}$, the emission matrix $\mathbf{E}$, and the second-order reaction matrix $\mathbf{R}_2$. Bold-italic (e.g. $\mathbf{J}^x$, $\mathbf{D}^x$, ∇ , σ) is reserved for ordinary vectors and tensors in *Cartesian space*. Thus, the entries $\mathbf{J}^x$ of the species-space flux array $\mathbf{J}$ are themselves Cartesian vectors, and the divergence $\nabla \cdot$ acts on each entry.

The stacked species flux collects the four Cartesian fluxes of the mobile species,

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}^v \\ \mathbf{J}^i \\ \mathbf{J}^{2i} \\ \mathbf{J}^{3i} \end{bmatrix}. \quad (2)$$

Here $\mathbf{J}^x = -\mathbf{D}^x \nabla c^x$ is the diffusive flux of species x , with a possibly-anisotropic diffusion tensor $D_{ij}^x = D_{0ij}^x \exp(-E_{ij}^{xm}/k_B T)$ whose pre-exponential factors D_{0ij}^x and migration energies E_{ij}^{xm} are the material-file vectors `mobileSpeciesD0_SI` and `mobileSpeciesEnergyMigration_eV`.

Two equivalent measures of mobility appear in what follows. The tensor $\mathbf{D}^x$ enters the transport term of Eq. (1) directly, whereas the reaction terms are written more compactly with the scalar jump frequency ω_x , the rate at which a defect of species x completes a migration jump. The two are the same quantity in different units,

$$\omega_x = z_c \nu_x \exp\left(-\frac{E_x^m}{k_B T}\right) = \frac{z_c}{a^2} \overline{D}^x, \quad \overline{D}^x = \frac{a^2}{z_c} \omega_x, \quad (3)$$

where ν_x is the attempt frequency, E_x^m the migration energy, z_c the combinatorial factor of the jump, a the lattice parameter, and $\overline{D}^x$ the orientationally averaged scalar diffusion coefficient obtained from $\mathbf{D}^x$. This is the relation the 0D model of Section 3.1 uses, so one set of migration energies fixes both the transport tensor and every reaction rate. The jump frequencies enter below in three places: the pairwise reaction rates $\beta^{m,n}$ of Eq. (9), the network sink term of Eq. (13), and the point-defect fluxes ϕ_v and ϕ_i that drive loop growth in Eq. (42).

The vector $\mathbf{P}_M$ is the cascade production term,

$$\mathbf{P}_M = G\varepsilon\mathbf{f}, \quad (4)$$

where G is the nominal dpa rate (`doseRate_dpaPerSec`), and ε is the cascade surviving efficiency (`mobileSpeciesSurvivingEfficiency`), and $\mathbf{f}$ the fraction vector partitioning cascade production among the mobile clusters¹.

The $\mathbf{R}_2$ -term represents second-order reactions between mobile species, modeling the collision of two mobile species to form a third species. The second-order reaction matrix, whose rows contract the concentration array with the per-species reaction matrices $\mathbf{R}_2^x$ of Eq. (7), is

$$\mathbf{R}_2 = \begin{bmatrix} \mathbf{R}_2^v \\ \mathbf{R}_2^i \\ \mathbf{R}_2^{2i} \\ \mathbf{R}_2^{3i} \end{bmatrix}. \quad (5)$$

In Eq. (1) this stacked array enters through the contraction $\frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M$, where, in the case of four mobile species, $\mathbf{I}$ is the 4×4 identity and $\otimes$ the Kronecker product:

$$\mathbf{I} \otimes \mathbf{c}_M^T = \begin{bmatrix} \mathbf{c}_M^T & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{c}_M^T & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{c}_M^T & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{c}_M^T \end{bmatrix}. \quad (6)$$

Note that the x -th row in (1) corresponding to the second-order reaction term is $\frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$, so this term is a vector in species space. For the chosen vector of mobile species, we have

$$\begin{aligned} \mathbf{R}_2^v &= - \begin{bmatrix} 0 & \beta^{v,i} & \beta^{v,2i} & \beta^{v,3i} \\ \beta^{v,i} & 0 & 0 & 0 \\ \beta^{v,2i} & 0 & 0 & 0 \\ \beta^{v,3i} & 0 & 0 & 0 \end{bmatrix}, & \mathbf{R}_2^i &= - \begin{bmatrix} 0 & \beta^{v,i} & -\beta^{v,2i} & 0 \\ \beta^{v,i} & 2\beta^{i,i} & \beta^{i,2i} & \beta^{i,3i} \\ -\beta^{v,2i} & \beta^{i,2i} & 0 & 0 \\ 0 & \beta^{i,3i} & 0 & 0 \end{bmatrix}, \\ \mathbf{R}_2^{2i} &= -2 \begin{bmatrix} 0 & 0 & \beta^{v,2i} & -\beta^{v,3i} \\ 0 & -\beta^{i,i} & \beta^{i,2i} & 0 \\ \beta^{v,2i} & \beta^{i,2i} & 2\beta^{2i,2i} & \beta^{2i,3i} \\ -\beta^{v,3i} & 0 & \beta^{2i,3i} & 0 \end{bmatrix}, & \mathbf{R}_2^{3i} &= -3 \begin{bmatrix} 0 & 0 & 0 & \beta^{v,3i} \\ 0 & 0 & -\beta^{i,2i} & \beta^{i,3i} \\ 0 & -\beta^{i,2i} & 0 & \beta^{2i,3i} \\ \beta^{v,3i} & \beta^{i,3i} & \beta^{2i,3i} & 2\beta^{3i,3i} \end{bmatrix}, \end{aligned} \quad (7)$$

where $\beta^{m,n}$ in Eq. (7) represents the reaction rates between species m and n . A positive reaction rate indicates production, while a negative sign means consumption. Each rate constant is built from the encounter geometry of the two reactants,

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega |\mathcal{M}_m \mathcal{M}_n|} (\overline{D}^m + \overline{D}^n) (r^m + r^n), \quad (8)$$

that is, the sum of the two reactant diffusivities weighted by their combined capture radius $r^m + r^n$, with $A^{m,n}$ a geometric prefactor and Ω the atomic volume. Substituting $\overline{D}^x = (a^2/z_c) \omega_x$ from Eq. (3) makes $\beta^{m,n}$ proportional to the sum of the two reactant jump frequencies, and the prefactors $A^{m,n}$ are chosen, through Eq. (119), so that the constant of proportionality is unity:

$$\beta^{m,n} = \omega_m + \omega_n. \quad (9)$$

¹Expanded component-wise, and accounting for the per-cluster entry of the di- and tri-interstitial sources ($G_{2i}/2$ and $G_{3i}/3$), $\mathbf{P}_M$ reproduces the 0D cascade sources $\{G_v, G_i, \frac{1}{2}G_{2i}, \frac{1}{3}G_{3i}\}$ of Section 3.1, with $G_x = G\varepsilon_x$ the corresponding partition of the displacement rate.

This is the form the 0D model of Section 3.1 uses for its pairwise reaction rates, $R_{a,b} = (\omega_a + \omega_b)C_aC_b$, and the two constructions agree to within 0.1 % on the assembled coefficients (Table 3). Two channels carry extra factors: the recombination-volume factor β_r on the interstitial-vacancy channel, $\beta^{v,i} \rightarrow \beta_r(\omega_i + \omega_v)$, and the combinatorial factors retained on the like-species reactions, $R_{i,i} = 2\omega_iC_i^2$ and $R_{2i,2i} = 4\omega_{2i}C_{2i}^2$. Expanding $\frac{1}{2}\mathbf{c}_M^T\mathbf{R}_M^x\mathbf{c}_M$ component-wise then reproduces the 0D pair-reaction budget of each monomer and small-cluster equation; the interstitial line, for instance, yields $R_{2i,v} + 2\mathcal{E}_{2i} + 3\mathcal{E}_{3i} - (R_{i,v} + R_{i,i} + R_{i,2i} + R_{i,3i})$ once the emission and sink terms below are added. The capture radius r^m used to build $\beta^{m,n}$ follows the point-defect / prismatic-loop geometry

$$r^m = \begin{cases} \left(\frac{3\Omega_0}{4\pi}\right)^{1/3}, & |m^m| = 1, \\ \left(\frac{|m^m|\Omega_0}{\pi b}\right)^{1/2}, & |m^m| > 1. \end{cases} \quad (10)$$

The $\mathbf{E}$ -term represents the thermal dissociation (emission) of mobile species. It is the matrix

$$\mathbf{E} = - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\alpha^{i,2i} & -\frac{1}{3}\alpha^{i,3i} \\ 0 & 0 & \alpha^{i,2i} & -\frac{2}{3}\alpha^{i,3i} \\ 0 & 0 & 0 & \alpha^{i,3i} \end{bmatrix}, \quad (11)$$

where $\alpha^{x,nx}$ is the dissociation rate of an nx -cluster emitting a point defect of type x ,

$$\alpha^{m,n} = \beta^{m,n} \exp\left(-\frac{E_{m,n}^b}{k_B T}\right), \quad (12)$$

With $E_{m,n}^b$, the binding energy of the $(m+n)$ -size cluster (`mobileSpeciesBindingEnergy_eV`). Component-wise, $\mathbf{E}\mathbf{c}_M$ reproduces the 0D thermal-emission terms $\mathcal{E}_{2i} = \omega_{2i}e_{2i}C_{2i}$ and $\mathcal{E}_{3i} = \omega_{3i}e_{3i}C_{3i}$ of Section 3.1.

The $\mathbf{D}$ -term is the linear reaction matrix that governs the annihilation of mobile defects at immobile sinks. It absorbs the mobile diffusivities into the sink strengths, which themselves combine the network sinks with the mass-conserving loop absorption. The mobile-to-immobile reaction matrix is:

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_{\mathcal{L}}, \quad \mathbf{D}_d = \text{diag}\{\bar{D}^v, \bar{D}^i, \bar{D}^{2i}, \bar{D}^{3i}\} \quad (13)$$

where $\mathbf{C}_s = [C_s^v, C_s^i, C_s^i, C_s^i]^T$ are the network-only sink concentrations of Section 3.1 ($C_s^v = \frac{a^2}{z_c}\rho_N$, $C_s^i = \frac{a^2}{z_c}Z_N\rho_N$) and $\boldsymbol{\omega} = [\omega_v, \omega_i, \omega_i, \omega_i]^T$. The first term reproduces the network sink reactions $R_{x,s} = \omega_x C_x C_s^x$, while $\mathbf{K}_{\mathcal{L}}$ carries the loop-absorption debits `loop_abs_i`, `loop_abs_v` defined in Eq. (43), so that mobile losses to loops match exactly the content gained by the immobile families and no defect is double-counted. The network term is already a reaction rate, whereas the loop sink strengths acquire the mobile diffusivities, so both contributions have units of frequency (Hz), and $\bar{D}^m$ is the (orientationally averaged) diffusion coefficient of mobile species m .

Here $\odot$ denotes the Hadamard (component-wise) product of two species-space vectors, so that the first term of Eq. (13) is the diagonal matrix

$$\text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} = \begin{bmatrix} \omega_v C_s^v & 0 & 0 & 0 \\ 0 & \omega_i C_s^i & 0 & 0 \\ 0 & 0 & \omega_i C_s^i & 0 \\ 0 & 0 & 0 & \omega_i C_s^i \end{bmatrix}. \quad (14)$$

Its x -th diagonal entry multiplies c^x in Eq. (1) and so returns the network sink reaction $R_{x,s} = \omega_x c^x C_s^x$ species by species. The three interstitial species share the monomer jump frequency ω_i and the interstitial sink concentration C_s^i , which is why the last three diagonal entries are equal.

The mobile species PDE (1) is accompanied by the concentration boundary condition² (BC)

$$\mathbf{c}_M(\mathbf{x}) = \mathbf{c}_{\text{bnd,eq}}(\mathbf{x}) \quad (\text{mobile species BC}), \quad (16)$$

²Eq. (16) is a pure essential (Dirichlet) BC representing a perfect sink. A more general Robin-type BC

$$-\mathbf{J}^k \cdot \mathbf{n} = h^k [c^k(\mathbf{x}) - c_{\text{bnd,eq}}^k(\mathbf{x})], \quad (15)$$

recovers the perfect sink condition as $h^k \rightarrow \infty$ and the no-flux condition as $h^k \rightarrow 0$.

where $c_{\text{bnd,eq}}^k(\mathbf{x})$ is the local concentration of the species in equilibrium with the boundary,

$$c_{\text{bnd,eq}}^k(\mathbf{x}) = c_0^k(\mathbf{x}) \exp\left(-\frac{m^k t_n(\mathbf{x}) \Omega}{k_B T}\right), \quad (17)$$

in which t_n is the normal traction on the boundary and $c_0^k(\mathbf{x})$ is the thermal-equilibrium concentration

$$c_0^k(\mathbf{x}) = |m^k| \exp\left(-\frac{E_k^f + m^k q \Omega_k^r}{k_B T}\right), \quad (18)$$

where $q(\mathbf{x}) = -\text{tr}(\boldsymbol{\sigma}(\mathbf{x}))/3$ is the local pressure, $\Omega_k^r = \omega_k^r \Omega$ the relaxation volume of the k -th cluster (with a relative relaxation volume ω_k^r , `mobileSpeciesRelRelaxVol`), and E_k^f its boundary formation energy (`mobileSpeciesEnergyFormation_eV`). The stress terms in Eqs. (17) and (18) are multiplied by the signed cluster size m^k , rendering opposite effects for interstitial and vacancy-type clusters.

2.3 Bias factors and loop sink strengths

The loop sink-strength block $\mathbf{K}_{\mathcal{L}}$ of Eq. (13) is diagonal in the mobile species and collects, for each mobile defect m , the total sink strength of the immobile populations,

$$\mathbf{K}_{\mathcal{L}} = \text{diag}\{\mathbf{k}^2\}, \quad \mathbf{k}^2 = \begin{bmatrix} k_v^2 \\ k_i^2 \\ k_{2i}^2 \\ k_{3i}^2 \end{bmatrix}, \quad k_m^2 = \sum_{k \in \mathcal{I}} \langle k_{k,m}^2 \rangle, \quad (19)$$

where $\mathcal{I} = \{vL, a_1L, a_2L, a_3L, bp\}$ indexes the immobile families of Section 2.4: the basal vacancy loop, the three prismatic interstitial loops, and the vacancy bi-pyramid. Every entry of $\mathbf{k}^2$ is a bias factor multiplied by a capture geometry. Both are defined once here, in index form over $\mathcal{I}$, and are used unchanged wherever they reappear.

Assign to each family $k \in \mathcal{I}$ a habit class $\chi(k) \in \{a, c\}$: prismatic ($\chi = a$) for the three interstitial loops, basal ($\chi = c$) for the vacancy loop and for the bi-pyramid, which flattens onto the basal plane. The bias factor Z_k^m is the capture efficiency of family k for mobile species m , and it carries two distinct effects. The first is the diffusional anisotropy difference (DAD): in the non-cubic h.c.p. lattice a defect does not sample all directions equally, so a loop is approached preferentially along one axis and its efficiency depends on its habit. Writing $p^m = (D_c^m/D_a^m)^{1/6}$ for the diffusional anisotropy of species m and Z_m^0 for its bias factor in the isotropic, stress-free limit, the stress-free capture efficiencies are

$$Z_{a,m}^{\text{DAD},0} = Z_m^0 \left(\frac{p^m + (p^m)^{-2}}{2} \right), \quad Z_{c,m}^{\text{DAD},0} = Z_m^0 p^m. \quad (20)$$

Unlike traditional cluster-dynamics models, which presume a fixed bias factor for DAD, the present spatially-resolved formulation obtains it from the tensorial description of the mobile-species flux: p^m follows from the same diffusion tensor $\mathbf{D}^m$ that enters the transport term of Eq. (1). The second effect is stress-induced preferential absorption (SIPA), which adds a correction $\Delta Z_{\chi(k),m}^{\text{SIPA}}$ evaluated at the stress the family resolves on its own habit. The bias factor of family k is therefore

$$Z_k^m(\boldsymbol{\sigma}) = Z_{\chi(k),m}^{\text{DAD},0} + \Delta Z_{\chi(k),m}^{\text{SIPA}}(\sigma_k), \quad \sigma_k = \begin{cases} \hat{\mathbf{b}}_k \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{b}}_k, & k \in \{a_1L, a_2L, a_3L\}, \\ \boldsymbol{\sigma}, & k \in \{vL, bp\}, \end{cases} \quad (21)$$

where $\hat{\mathbf{b}}_k$ is the unit Burgers direction of family k , so that the three prismatic loops resolve σ_{11} , σ_{22} and σ_{33} along a_1 , a_2 and a_3 , while the basal families, whose habit is normal to $[0001]$, take the stress state as a whole. Equation (21) is the only definition of the bias factors in this work: the same Z_k^m drive the loop growth of Eq. (43) and the mobile-defect sinks assembled here, so the bias enters the model exactly once and is not restated where it is used.

The sink strength of family k for species m is that bias factor multiplied by the family's capture geometry,

$$\langle k_{k,m}^2 \rangle = Z_k^m s_k \pi r^k n^k, \quad s_k = \begin{cases} 2, & k \in \{vL, a_1L, a_2L, a_3L\}, \\ \alpha_{bp} s_{bp}, & k = bp, \end{cases} \quad (22)$$

with r^k the mean radius of the family and n^k its number density. For the four loop families $s_k \pi r^k = 2\pi r^k$ is the loop circumference, the line length on which a diffusing defect can be captured; for the bi-pyramid the shape factor s_{bp} and the efficiency α_{bp} take the place of that circumference.

At zero stress the SIPA term vanishes, $Z_k^m \rightarrow Z_{\chi(k),m}^{\text{DAD},0}$, and Eq. (22) reduces to $\langle k_{a_j L, m}^2 \rangle = Z_m^0 (p^m + (p^m)^{-2}) \pi r^{a_j L} n^{a_j L}$ for the prismatic loops and $\langle k_{v L, m}^2 \rangle = 2Z_m^0 p^m \pi r^{v L} n^{v L}$ for the basal loop. For an isotropically diffusing defect ($p^m = 1$) both recover the classical dislocation sink strength $2\pi Z_m^0 r n$. The di- and tri-interstitials migrate preferentially in the basal plane, so $p^m < 1$ and the basal loops are the weaker sink for them, $\langle k_{v L, m}^2 \rangle < \langle k_{a_j L, m}^2 \rangle$. The sink strength of the vacancy clusters is interpolated from the bi-pyramid value $\langle k_{bp, m}^2 \rangle$ toward the basal-loop value $\langle k_{v L, m}^2 \rangle$ as the reconstructed vacancy loops flatten onto the basal plane.

2.4 Immobile defect clusters

In the CD implementation of MoDELlib2, interstitial clusters larger than $m_{\max}$ and vacancy clusters larger than $m_{\min}$ are considered immobile and are treated as dislocation loops. In irradiated Zr four immobile loop families are represented, following Section 3.1: one vacancy ($\langle c \rangle$) loop family and three interstitial ($\langle a \rangle$) loop families indexed by the orientation of their Burgers vector along the three prismatic directions a_1, a_2, a_3 ,

$$\mathcal{L} = \{vL, a_1L, a_2L, a_3L\}, \quad (23)$$

Each family k carries a polarity $p^k \in \{+1, -1\}$ ($p = +1$ interstitial, $p = -1$ vacancy),

$$p^{vL} = -1, \quad p^{a_1L} = p^{a_2L} = p^{a_3L} = +1, \quad (24)$$

so that the interstitial and vacancy families are selected by the boolean vectors $\mathcal{I}_i = \frac{1}{2}(1 + p)$ and $\mathcal{I}_v = \frac{1}{2}(1 - p)$.

Each immobile family is described by two spatially-resolved fields: the *cluster density* n^k , i.e. the number of loops of family k per host atom, and the *defect content* c^k , i.e. the number of point defects stored in that family per host atom,

$$\mathbf{n}_I(\mathbf{x}) = [n^{vL}, n^{a_1L}, n^{a_2L}, n^{a_3L}]^T \quad (\text{cluster density, clusters per atom}) \quad (25)$$

$$\mathbf{c}_I(\mathbf{x}) = [c^{vL}, c^{a_1L}, c^{a_2L}, c^{a_3L}]^T \quad (\text{defect content, defects per atom}). \quad (26)$$

Both are dimensionless (fractional) quantities. The mean number of point defects per cluster of family k follows from the ratio of content to density,

$$m^k = \frac{c^k}{n^k}, \quad (27)$$

and the mean loop radius is recovered from the content-to-density ratio through the loop length scales ℓ_a, ℓ_c of Section 3.1,

$$r^k = \ell^k \sqrt{\frac{c^k}{n^k}}, \quad \ell^{vL} = \ell_c, \quad \ell^{a_1L} = \ell^{a_2L} = \ell^{a_3L} = \ell_a, \quad (28)$$

which, expanded family by family, reproduces the 0D content relations $r_{vL} = \ell_c \sqrt{C_{vL}^v / C_{vL}}$, $r_{a_1L} = \ell_a \sqrt{C_{a_1L}^i / C_{a_1L}}$ of Eq. (60)'s companion set, with the identification $n^k \leftrightarrow C_{XL}$ and $c^k \leftrightarrow C_{XL}^x$.

The shape of a cluster depends on its size m^k : it has been shown that vacancy clusters transition from a compact bi-pyramid geometry to a planar (platelet) loop with increasing cluster size. We therefore include this transition of clusters from volumetric defects to planar loops, so that the effective radius of family k interpolates between the two limiting geometries,

$$R^k(m^k) = [1 - S^k(m^k)] R_{\text{pyr}}^k + S^k(m^k) R_{\text{loop}}^k, \quad (29)$$

where the pyramid and loop radii are

$$R_{\text{pyr}}^k = \left(\frac{m^k \Omega_0}{\sqrt{8}} \right)^{1/3}, \quad R_{\text{loop}}^k = \sqrt{\frac{m^k \Omega_0}{\pi b^k}}. \quad (30)$$

The function S^k is a sigmoidal transition function operating a smooth switch between the two shapes,

$$S^k(m^k) = \frac{1}{1 + \exp \left[-\frac{m^k - m_0^k}{w^k} \right]}, \quad (31)$$

whose midpoint and steepness are set by

$$m_0^k = \frac{m_{\min}^k + m_{\max}^k}{2} - m_s^k, \quad w^k = w_0 (m_{\max}^k - m_{\min}^k), \quad (32)$$

so that clusters with $m^k \lesssim m_{\min}^k$ are predominantly bi-pyramid-like while clusters with $m^k \gtrsim m_{\max}^k$ are predominantly loop-like. The same sigmoidal function interpolates the cluster formation-volume tensor,

$$\Omega^k(m^k) = (1 - S^k(m^k)) \Omega_{\text{pyr}}^k + S^k(m^k) \Omega_{\text{loop}}^k, \quad (33)$$

with the loop and pyramid limits

$$\Omega_{\text{loop}}^k = \mathcal{I}^k \frac{\mathbf{b}^k \otimes \mathbf{b}^k}{\mathbf{b}^k \cdot \mathbf{b}^k} V^k, \quad \Omega_{\text{pyr}}^k = \mathcal{I}^k \frac{\mathbb{I}}{3} V^k, \quad (34)$$

where $V^k = m^k \Omega_0$ is the cluster volume and $\mathcal{I}^k$ the loop selector. The pyramid regime applies to the sub-critical vacancy ($\langle c \rangle$) family, which collapses smoothly onto the planar vacancy loop as it grows.

Both fields obey spatially-resolved ordinary differential equations coupled to the mobile PDE (1): a *density* equation for $\mathbf{n}_I$ (nucleation and annealing of loops) and a *content* equation for $\mathbf{c}_I$ (defect storage through absorption). These are developed in turn below.

The loop density evolves by nucleation and thermal annealing,

$$\frac{d\mathbf{n}_I}{dt} = \dot{\mathbf{N}}^{\text{nuc}} - \mathbf{a}_n - \Delta \mathbf{N}, \quad (35)$$

where $\dot{\mathbf{N}}^{\text{nuc}}$ is the nucleation-rate vector, $\mathbf{a}_n$ the (vacancy-loop) annealing vector, and $\Delta \mathbf{N}$ the coalescence sink. The coalescence sink is diagonal in the loop family and, for each family $k \in \{vL, a_1L, a_2L, a_3L\}$, reproduces the two-channel geometric coalescence of Section 3.1 (a like-loop channel that conserves content and a loop-network channel that is booked as network absorption),

$$\Delta \mathbf{N} = [\Delta N^k]_{k \in \mathcal{L}}, \quad \Delta N^k = (\nu_{LL}^k \phi_{LL}^k + \nu_{LN}^k \phi_{LN}^k) n^k, \quad (36)$$

with the Avrami overlap probabilities ϕ_{LL}^k, ϕ_{LN}^k and rate scales ν_{LL}^k, ν_{LN}^k of the coalescence model of Section 3.1, and n^k the family density. The two terms are exactly the 0D like-loop and loop-network number-loss rates $\Delta N = \nu_{LL} \phi_{LL} n + \Delta N_{LN}$ with the 0D number concentration replaced family by family by n^k . Interstitial loops nucleate from homogeneous clustering of mobile species plus a cascade seed, while vacancy loops nucleate from the cascade only,

$$\dot{\mathbf{N}}^{\text{nuc}} = \mathbf{w}(\boldsymbol{\sigma}) \Phi_{\text{clus}} + \mathbf{G}_{\mathcal{L}} \oslash \mathbf{n}^{\text{nuc}}, \quad (37)$$

with $\Phi_{\text{clus}} = R_{i,3i} + R_{2i,2i}$ the interstitial-clustering flux, the stress-dependent orientation weight vector $\mathbf{w}(\boldsymbol{\sigma}) = [0, w_{a_1}(\boldsymbol{\sigma}), w_{a_2}(\boldsymbol{\sigma}), w_{a_3}(\boldsymbol{\sigma})]^T$, the cascade-loop source vector $\mathbf{G}_{\mathcal{L}} = [G_{vL}, G_{a_1L}, G_{a_2L}, G_{a_3L}]^T$, per-loop seed sizes $\mathbf{n}^{\text{nuc}} = [n_{vL}^{\text{nuc}}, n_{ia}^{\text{nuc}}, n_{ia}^{\text{nuc}}, n_{ia}^{\text{nuc}}]^T$, and $\oslash$ the elementwise (Hadamard) division. Expanding (37) component-wise recovers exactly the 0D nucleation rates $\dot{N}_{a_1L}^{\text{nuc}} = w_{a_1}(\boldsymbol{\sigma})(R_{i,3i} + R_{2i,2i}) + G_{a_1L}/n_{ia}^{\text{nuc}}$, and likewise for a_2L, a_3L, vL . The orientation weights $w_{a_k}(\boldsymbol{\sigma}) = \frac{1}{3}(1 + \chi \hat{\sigma}_{kk})$, with $\sum_k w_{a_k} = 1$, bias the cascade nucleation seed among the three prismatic directions through the resolved stress σ_{kk} of Eq. (21); at zero stress they reduce to the common seed $\frac{1}{3}$. These linearized weights are the small-stress limit of the stress-induced preferential nucleation (SIPN) partition, in

which the cascade-produced defects are distributed among the immobile families through a Boltzmann factor coupling the applied stress $\boldsymbol{\sigma}$ to each family's formation-volume tensor $\boldsymbol{\Omega}^k$,

$$\chi_i^k = \frac{\exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^k}{k_B T}\right) \mathcal{I}_i^k}{\sum_p \exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^p}{k_B T}\right) \mathcal{I}_i^p}, \quad \chi_v^k = \frac{\exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^k}{k_B T}\right) \mathcal{I}_v^k}{\sum_p \exp\left(\frac{\boldsymbol{\sigma} : \boldsymbol{\Omega}^p}{k_B T}\right) \mathcal{I}_v^p}, \quad (38)$$

where $\mathcal{I}_i^k$ and $\mathcal{I}_v^k$ are the interstitial and vacancy selectors, and the sums run over all immobile families p . At zero stress every $\boldsymbol{\sigma} : \boldsymbol{\Omega}^k$ vanishes and the interstitial partition reduces to the common seed $\frac{1}{3}$ among the three prismatic orientations, recovering $w_{a_k}(\mathbf{0}) = \frac{1}{3}$. The annealing vector acts only on the vacancy families,

$$\mathbf{a}_n = \left[\frac{n^{vL}}{\tau_{vL}}, 0, 0, 0 \right]^T, \quad (39)$$

with the Arrhenius lifetimes $\tau_{vL}(T) = \tau_{vL}^0 \exp(E_a^{vL}/k_B T)$.

The defect content of the loops evolves through mass-conserving absorption of mobile defects, cascade deposition, nucleation seeding, and coalescence,

$$\frac{d\mathbf{C}_I}{dt} = \dot{\mathbf{G}}^c + \mathbf{P}_G + \mathbf{s}^{\text{nuc}} - \mathbf{A}_c - \Delta\mathbf{C}. \quad (40)$$

The coalescence content sink removes stored content only through the loop-network channel (the like-loop channel merges two loops without losing content), so, family by family,

$$\Delta\mathbf{C} = [\Delta C^k]_{k \in \mathcal{L}}, \quad \Delta C^k = \nu_{LN}^k \phi_{LN}^k c^k, \quad (41)$$

with c^k the family content, matching the 0D content-loss rate $\Delta C = \nu_{LN} \phi_{LN} c$ of Section 3.1 with the 0D content replaced family by family by c^k . The net content-growth vector $\dot{\mathbf{G}}^c$ collects, for each family, the absorption of its own defect type minus absorption of the opposite type (a defect-loop recombination). The mobile point-defect fluxes are the vacancy flux ϕ_v and the interstitial flux ϕ_i ,

$$\phi_v = \omega_v c^v, \quad \phi_i = \omega_i (c^i + 2c^{2i} + 3c^{3i}), \quad (42)$$

from Section 3.1. Each immobile family grows by capturing its like defect and shrinks by capturing the opposite defect: the vacancy loop vL captures ϕ_v and recombines with ϕ_i , whereas the three prismatic interstitial loops a_1L , a_2L and a_3L capture ϕ_i and recombine with ϕ_v . With the geometric capture prefactor $\pi_k = (\ell_c/\ell)\sqrt{c^k n^k}$ of family k (with the c -loop efficiency Q folded into π_{vL}) and the orientation-resolved capture efficiencies Z^k , the absorption prefactors and the diffusional-anisotropy-difference (DAD) bias give the content-growth rate of each family,

$$\dot{G}_{vL}^c = p_{vL} \pi_{vL} (Z_v^{vL} \phi_v - Z_i^{vL} \phi_i g(r_{vL})), \quad (43)$$

$$\dot{G}_{a_1L}^c = p_{a_1L} \pi_{a_1L} (Z_i^{a_1L} \phi_i - Z_v^{a_1L} \phi_v), \quad (44)$$

$$\dot{G}_{a_2L}^c = p_{a_2L} \pi_{a_2L} (Z_i^{a_2L} \phi_i - Z_v^{a_2L} \phi_v), \quad (45)$$

$$\dot{G}_{a_3L}^c = p_{a_3L} \pi_{a_3L} (Z_i^{a_3L} \phi_i - Z_v^{a_3L} \phi_v), \quad (46)$$

where p_k is the defect sign of family k ($p = -1$ for the vacancy loop, $p = +1$ for the interstitial loops) and the minimum-stable-size gate $g(r_{vL})$ acts only on the vacancy-loop channel. The capture efficiencies Z_m^k are those of Eq. (21), evaluated family by family: each interstitial loop a_kL carries the prismatic bias at its own resolved stress σ_{kk} , while the vacancy loop carries the basal (c -type) bias. All of them reduce to their stress-free DAD values $Z_{a,m}^{\text{DAD},0}$ (interstitial loops) and $Z_{c,m}^{\text{DAD},0}$ (vacancy loop) when $\boldsymbol{\sigma} = 0$. Collected in the vacancy-first order $\mathcal{L} = \{vL, a_1L, a_2L, a_3L\}$, these four expressions are the entries of the content-growth vector $\dot{\mathbf{G}}^c = [\dot{G}_k^c]_{k \in \mathcal{L}}$ of Eq. (40).

The cascade deposition vector $\mathbf{P}_G$ delivers the loop-borne fraction of the collision cascade,

$$\mathbf{P}_G = \mathbf{G}_\mathcal{L} = G(\varepsilon_{ia} \mathbf{w}(\boldsymbol{\sigma}) + \varepsilon_{vL} \mathbf{w}'), \quad (47)$$

with $\mathbf{w}(\boldsymbol{\sigma}) = [0, w_{a_1}, w_{a_2}, w_{a_3}]^T$ and $\mathbf{w}' = [1, 0, 0, 0]^T$, so that the stress-biased partition among the three interstitial orientations and the vacancy loop is preserved. The nucleation-seeding vector deposits the interstitial atoms consumed by the clustering reactions as newborn loop content,

$$\mathbf{s}^{\text{nuc}} = \mathcal{N}_c [0, w_{a_1}, w_{a_2}, w_{a_3}]^T, \quad \mathcal{N}_c = 4R_{i,3i} + 2R_{2i,2i} + 3R_{2i,3i}, \quad (48)$$

and the content-annealing vector returns the birth content of dissolving vacancy embryos to the free pool,

$$\mathbf{A}_c = \left[\frac{c^{vL}}{\tau_{vL}}, 0, 0, 0 \right]^T. \quad (49)$$

The lifetime τ_{vL} above is the size-independent survival lifetime of the sub-critical vacancy embryos produced in the cascade, so that $\mathbf{a}_n$ and $\mathbf{A}_c$ describe the *dissolution* of embryos that fail to reach the minimum stable size; the minimum-stable-size gate $g(r^{vL})$ of (43) confines these terms to the sub-critical population. Vacancy loops that survive and grow lose vacancies through a second, distinct channel: the *thermal emission* of single vacancies from a stable loop, governed by the size-dependent vacancy binding energy. Because the reduced model tracks only the family density n^k and content c^k , the emission rate is evaluated at the mean loop size

$$\bar{m}^k = \frac{c^k}{n^k}, \quad (50)$$

and, following the detailed-balance form of (9)–(11), the per-defect thermal-emission rate of the vacancy loop is

$$\alpha^{v,k}(\bar{m}^k) = \beta_{v,k} \exp\left(-\frac{E_b^{v,k}(\bar{m}^k)}{k_B T}\right), \quad k = vL, \quad (51)$$

where $E_b^{v,k}(\bar{m}^k)$ is the vacancy binding energy of the mean-size loop, which grows toward its bulk limit with $\bar{m}^k$ so that larger loops emit ever more weakly. This channel reduces the loop content without removing loops, and hence enters the content ODE (40) through the thermal-emission vector

$$\mathbf{A}_c^{\text{emit}} = [\alpha^{v,vL}(\bar{m}^{vL}) c^{vL}, 0, 0, 0]^T, \quad (52)$$

which is subtracted alongside $\mathbf{A}_c$ while leaving the density ODE unchanged.

The vacancies released by *both* channels are returned to the free vacancy pool, sourcing the mobile species PDE (1) through

$$\mathbf{S}_{vL} = \left[\frac{c^{vL}}{\tau_{vL}} + \alpha^{v,vL}(\bar{m}^{vL}) c^{vL}, 0, 0, 0 \right]^T, \quad (53)$$

whose single non-zero entry feeds the mobile vacancy line. By construction $\mathbf{S}_{vL}$ is term-by-term identical to the vacancy content removed from the loops by $\mathbf{A}_c + \mathbf{A}_c^{\text{emit}}$, so the exchange is a pure transfer: every vacancy debited from the immobile families is credited to the mobile pool exactly once, and the total point-defect content is conserved. This transfer is independent of the mass-conserving absorption coupling $\dot{\mathbf{G}}^c$, so a surviving loop may simultaneously grow by absorbing the excess vacancy flux (debited from $\mathbf{c}_M$ via $\mathbf{D}$) and shrink by thermal emission (credited to $\mathbf{c}_M$ via $\mathbf{S}_{vL}$) without any double counting. Expanding (40) family by family reproduces the 0D content equations, e.g. $dC_{a_1L}^i/dt = \dot{G}_{a_1L}^c + w_{a_1}\mathcal{N}_c + G_{a_1L} - \Delta C_{a_1L}$ and $dC_{vL}^v/dt = \dot{G}_{vL}^c + G_{vL} - \text{ann}_{vL}^c - \Delta C_{vL}$. The corresponding debits enter the mobile PDE (1) through the reaction term $\mathbf{D}\mathbf{c}_M$, with $\text{loop_abs}_i = \sum_k [\dot{G}_k^c]_i$ and $\text{loop_abs}_v = \sum_k [\dot{G}_k^c]_v$, so no defect is double-counted.

3 The ZrMicro Code

Two codes implement the model of Section 2. **ZrMicro-0d** solves the classical rate-theory description of microstructure evolution in neutron-irradiated zirconium as a system of ordinary differential equations for spatially uniform concentrations. **ZrMicro-3d** is a branch of **MoDeLib2-NNL** that solves the

same reaction kinetics on a finite-element mesh, carrying the mobile species as resolved fields and the immobile loop populations as continuum densities and contents. They were developed in that order, and deliberately so.

The reduced model comes first because it is the cheapest place to establish that the physics is right. It integrates in seconds over eight decades of dose, so every mechanism — the cascade partition, the diffusional-anisotropy-difference bias, geometric coalescence, embryo dissolution, the evolving network density — can be switched on individually and its consequences followed to saturation. The equivalent three-dimensional calculation takes hours and confounds a modelling error with a discretization or boundary artifact, so a mechanism that has not first been exercised in 0D is difficult to diagnose once it is embedded in a spatial solve.

The second role follows from the first and is the more important one. The parameters of the model — cascade clustering efficiencies, loop nucleation sizes, coalescence coefficients, DAD bias factors, the vacancy-loop dissolution lifetime — are not independent first-principles quantities. They are fitted jointly against measured $\langle a \rangle$ and $\langle c \rangle$ loop number densities and mean sizes as functions of dose and temperature (Section 4), and the predictive content of the model resides in that calibration. A fit of this kind requires thousands of forward solves, which is affordable in 0D and is not affordable in 3D. The reduced model is therefore used as a *cheap surrogate* for the spatial one: the parameter set is identified in **ZrMicro-0d** and transferred to the **ZrMicro-3d** material file.

That transfer is legitimate only if the two codes integrate the same model, and two distinct things stood in the way. The first is structural and was resolved by the two-stage development described in Section 2.1: **MoDElib2-NNL** as distributed compiles with `iSize = 0` and carries no immobile population, so the loop families had to be written into it, first as content equations alone at `iSize = 4` with the density frozen, then at `iSize = 8` with the density promoted to a state variable and the nucleation, coalescence and dissolution kinetics that a free density makes meaningful. Until that was done there was nothing in the spatial code for the 0D calibration to be transferred *to*.

The second obstacle is parametric and remains even now that the equations agree. **MoDElib2-NNL** carries its own, independent parameterization, and it disagrees with **ZrMicro-0d** on the surviving fraction of cascade defects, on the migration energies, on the atomic volume and on the reaction rate coefficients. A 3D calculation performed with those values would be a different physical model however faithfully the loop equations were implemented, and the experimental calibration would not transfer to it. The reconciliation therefore has a precise objective:

*In the limit of a domain large compared with the point-defect depletion length, the interior solution of the 3D code must reproduce the 0D solution of **ZrMicro-0d** to within the differences that are deliberate and documented.*

Only when this is demonstrated can spatial effects computed by the 3D code be interpreted as physics rather than as an artifact of an inconsistent parameter set. The demonstration is deferred to Section 5.2; what follows here is the model each code integrates and the mapping between them. The test that matters is the dose dependence rather than the agreement at any one dose: a residual error in units or in a rate constant almost always produces a discrepancy that *grows* with dose, because the loop populations integrate the mobile fluxes, so a ratio that stays constant in dose is strong evidence that the remaining differences are structural and understood rather than accidental.

The two architectures differ in ways that matter for the comparison. **ZrMicro-0d** advances a state vector of nineteen unknowns — twelve physical concentrations, six conservation accumulators and the network dislocation density ρ_N — with a single stiff BDF integrator; there is no transport, and all nineteen equations are integrated together in time. **ZrMicro-3d** instead treats the two families on their two natural time scales. The mobile species are fast: their relaxation time to the local sink field is of order milliseconds, against dose steps of order 10^7 s, so they are solved as the quasi-steady reaction-diffusion problem of Eq. (1), which carries no time derivative at all, by Newton iteration on a second-order tetrahedral finite-element space. The immobile species carry no transport and are marched pointwise in dose at each finite-element node. This operator split is the essential structural difference between the two codes.

Beneath it lie three differences of formulation, collected in Table 1. **ZrMicro-0d** works in jump frequencies ω_m where **ZrMicro-3d** works in diffusivities D_m , so every rate coefficient must be mapped rather than transcribed; **ZrMicro-0d** represents sinks through equivalent *sink concentrations* rather than sink strengths, which is the same $k^2 D c$ form written in different variables; and **ZrMicro-0d**

handles loop absorption explicitly and separately from the point-defect sink strength, to avoid counting it twice.

Table 1: Structural differences between the two formulations.

	ZrMicro-0d	ZrMicro-3d
Spatial resolution	none (0-D)	second-order FEM, 3-D
Mobile species	ODEs in time	steady-state PDE (QSSA)
Immobile species	ODEs in time	pointwise ODEs per node
Primitive rate variable	jump frequency ω_m	diffusivity D_m
Sinks	sink concentrations, Eq. (62)	sink strengths k^2 , Eq. (19)
Loop absorption	explicit, Eq. (76)	sink-strength term in Eq. (1)
Grain boundary	lumped sink term	Dirichlet condition on the mesh boundary
Loop orientation	aligned / non-aligned (SIPN)	one family per Burgers vector; separated only by resolved stress
Atomic volume	$\Omega = 1.2 \times 10^{-29} \text{ m}^3$	$\det(\text{lattice basis}) b^3$

Two entries in Table 1 deserve emphasis because they are the places where the 3D model adds physics rather than resolution. The first is the grain boundary. In **ZrMicro-0d** it is one more term in a lumped sink strength; in **ZrMicro-3d** it is a Dirichlet boundary condition on the mobile species, imposed at the thermal equilibrium concentration. That is the physics the spatial model adds, and it is the reason the loop-absorption sink in the 3D code must *not* include a grain-boundary contribution: doing so would count the same absorption twice. The second is the way loop orientation is carried. **ZrMicro-0d** splits each loop type into aligned and non-aligned populations weighted by the stress-induced preferential nucleation factors f_a and $1 - f_a$, whereas **ZrMicro-3d** indexes the three prismatic $\langle a \rangle$ families by their Burgers vector alongside one basal $\langle c \rangle$ family. The distinction is one of bookkeeping rather than of physics, because the three prismatic variants are crystallographically equivalent: nothing in the model separates them except the stress each resolves on its own Burgers direction, through the nucleation weights $w_{a_k}(\sigma)$ and the SIPA correction of Eq. (21). What distinguishes a variant is therefore whether it is aligned with the applied stress, exactly as in **ZrMicro-0d**; the crystallographic index is the means by which that alignment is expressed. At zero stress the two descriptions coincide identically, each prismatic family carrying one third of the total, and that gives a useful internal check: at zero stress the three $\langle a \rangle$ family densities must be equal, and any departure indicates an error in the crystallography rather than in the kinetics.

3.1 ZrMicro-0d

The reduced model is a development of the Li & Ghoniem (2020) cluster-dynamics formulation for irradiated zirconium; Table 2 summarizes where the implemented solver departs from it, and the equations below give each item in full.

The twelve physical species of the state vector are

$$\mathbf{y} = [C_v, C_i, C_{2i}, C_{3i}, C_{iL}, C_{aiL}, C_{vL}, C_{avL}, C_{iL}^i, C_{aiL}^i, C_{vL}^v, C_{avL}^v], \quad (54)$$

all expressed as atomic fractions: the first four are the mobile point defects and small interstitial clusters, the next four are loop number densities split into aligned and non-aligned families by the SIPN weights, and the last four are the corresponding stored defect contents. Six conservation accumulators and ρ_N complete the nineteen.

All rates are built from Arrhenius jump frequencies in the form of Eq. (3). Unlike the two-frequency original, the implemented model carries *four*, because the small interstitial clusters are given their own migration energy:

$$\omega_i = z_c \nu_i \exp\left(-\frac{E_i^m}{k_B T}\right), \quad \omega_{2i} = z_c \nu_i \exp\left(-\frac{E_{2i}^m}{k_B T}\right), \quad (55)$$

$$\omega_{3i} = \omega_{2i}, \quad \omega_v = z_c \nu_v \exp\left(-\frac{E_v^m}{k_B T}\right), \quad (56)$$

Table 2: Updates of the implemented code relative to the original Li–Ghoniem appendix. “LG” = original formulation; “Code” = current solver (`rate_equations.cpp` / `reaction_rates.py` / `input_data.py`).

Aspect	Li–Ghoniem (original)	ZrMicro-0d Code (implemented)
State dimension	12 species	19: 12 species + 6 conservation accumulators + evolving ρ_N
Jump frequencies	two (ω_i, ω_v)	four: ω_i, ω_{2i} (own E_{2i}^m), $\omega_{3i} = \omega_{2i}, \omega_v$
Reaction-rate constant	single (larger) frequency	sum ($\omega_a + \omega_b$); recombination $\times \beta_r$ (recom)
Loop capture bias	generic Z_i^{XL}, Z_v^{XL}	DAD scheme $Z_i^a=1+\delta_i, Z_v^a=1-\delta_v, Z_i^c=1-\delta_i, Z_v^c=1+\delta_v; \delta_i(T)$ linear
Point-defect sinks	network + loops + precipitates + grain boundary in sink strength	network-only; loop absorption split into explicit mass-conserving coupling
Loop binding energy	elastic-force $F_{el}(r)$ + stacking fault	constant 5 eV surrogate
dn_{vL}/dt	$G_{vL}(n_{vL}/c_{vL}^v)(1 - n_{cap}n_{vL}) - n_{vL}/\tau_{vL}$	$G_{vL}/n_{vL}^{nuc} - n_{vL}/\tau_{vL} - \Delta N_{vL}$ (no capture factor)
Loop nucleation	homogeneous clustering only	clustering + cascade source G_{iL}/n_{iL}^{nuc} (per-atom content seed)
Loop lifetime	cascade estimate $-r_{vL}(0)/\dot{r}_{vL} _0$	Arrhenius $\tau_{vL} = \tau_{vL}^0 \exp(E_a^{vL}/k_B T)$
Network density	fixed ρ_N	T -dependent seed (log-linear) + evolving state variable
New physics	—	geometric loop coalescence (like-loop + loop-network, Avrami gates); a-loop minimum-size gate; annealing content release; conservation accumulators

with $z_c = 48$ the combinatorial number and ν_i, ν_v the attempt frequencies (Debye-scale, $\sim 5 \times 10^{12}$ Hz, for Zr), whence $D_v = (a^2/z_c)\omega_v$, $D_i = (a^2/z_c)\omega_i$ and the thermal-equilibrium vacancy concentration $C_v^e = \exp(-E_v^f/k_B T)$. The Boltzmann emission probabilities are

$$e_v = \exp\left(-\frac{E_v^f}{k_B T}\right), \quad e_{2i} = \exp\left(-\frac{E_{2i}^B}{k_B T}\right), \quad e_{3i} = \exp\left(-\frac{E_{3i}^B}{k_B T}\right), \quad (57)$$

and the loop binding energies are presently taken as a single constant $E_{loop}^B \approx 5$ eV,

$$e_{v,iL} = e_{v,vL} = \exp\left(-\frac{5 \text{ eV}}{k_B T}\right), \quad e_{v,X}^\sigma = \exp\left(\frac{\sigma_H \Omega}{k_B T}\right) e_{v,X}, \quad X \in \{iL, vL, avL\}, \quad (58)$$

where the hydrostatic-stress factor multiplies the loop emission probabilities, σ_H being the hydrostatic stress and Ω the atomic volume. The elaborate elastic-force and stacking-fault loop binding energies of the original appendix, built on $F_{el}(r) = \mu b^2 \ln[(r+b)/b]/[4\pi(1-\nu)(r+b)]$, are not used in the current solver; they are replaced by this constant surrogate pending a refit.

In h.c.p. Zr the interstitial diffusion tensor is far more anisotropic (near-two dimensional, basal) than the vacancy tensor, and that *anisotropy difference* partitions arriving point defects by sink orientation, so that *a*-type (interstitial, prismatic) loops see a net interstitial flux while *c*-type (vacancy, basal) loops see a net vacancy flux, even at equal production and equal geometric capture. Where the 3D code obtains this bias from the diffusion tensor itself through Eq. (20), the reduced model carries it phenomenologically as a pair of orientation- and species-resolved capture efficiencies,

$$a\text{-loops } (iL, aiL): \quad Z_{a\mathcal{L},i}^{\text{DAD}} = 1 + \delta_i, \quad Z_{a\mathcal{L},v}^{\text{DAD}} = 1 - \delta_v, \quad (59)$$

$$c\text{-loops } (vL, avL): \quad Z_{c\mathcal{L},i}^{\text{DAD}} = 1 - \delta_i, \quad Z_{c\mathcal{L},v}^{\text{DAD}} = 1 + \delta_v, \quad (60)$$

where $\delta_v = \delta_{\text{DAD}}$ and the interstitial anisotropy δ_i is a temperature-linear interpolation between anchors at 300 and 600 K, clamped to (0, 1),

$$\delta_i(T) = \text{clip}\left[\delta_i^{300} + (\delta_i^{600} - \delta_i^{300}) \frac{T - 300}{600 - 300}, 10^{-3}, 0.95\right]. \quad (61)$$

With $x \equiv \phi_v/\phi_i$ the arrival ratio, a -loops grow for $x < (1 + \delta_i)/(1 - \delta_v)$ and c -loops for $x > (1 - \delta_i)/(1 + \delta_v)$; the two windows overlap for any $\delta > 0$, so both families can grow simultaneously.

The point-defect sink concentrations are network-only, loop absorption being handled separately by the mass-conserving coupling below, so that loops must not also appear here:

$$C_s^v = \frac{a^2}{z_c} \rho_N, \quad C_s^i = \frac{a^2}{z_c} Z_N \rho_N, \quad (62)$$

with Z_N the network interstitial bias and ρ_N the evolving network density. The total dislocation density $\rho_{tot} = \rho_N + \rho_{iL} + \rho_{aiL} + \rho_{vL} + \rho_{avL}$, with $\rho_{XL} = 2\pi r_{XL} C_{XL}/\Omega$, is retained for diagnostics only. The precipitate term $4\pi R_p^e N_p$ and the grain-boundary sink C_{GB} of the original appendix are not active in the current right-hand side. The point-defect fluxes that drive the loop bias are

$$\phi_v = \omega_v C_v, \quad \phi_i = \omega_i (C_i + 2C_{2i} + 3C_{3i}). \quad (63)$$

Reaction rates use the sum of the two reactant frequencies in the form $R_{a,b} = (\omega_a + \omega_b) C_a C_b$ established in Eq. (9), the original having used the larger frequency alone; interstitial-vacancy recombination additionally carries the recombination-volume factor β_r :

$$R_{i,v} = \beta_r (\omega_i + \omega_v) C_i C_v, \quad R_{2i,v} = (\omega_{2i} + \omega_v) C_{2i} C_v, \quad (64)$$

$$R_{3i,v} = (\omega_{3i} + \omega_v) C_{3i} C_v, \quad R_{i,i} = 2\omega_i C_i^2, \quad (65)$$

$$R_{i,2i} = (\omega_i + \omega_{2i}) C_i C_{2i}, \quad R_{i,3i} = (\omega_i + \omega_{3i}) C_i C_{3i}, \quad (66)$$

$$R_{2i,2i} = 4\omega_{2i} C_{2i}^2, \quad R_{2i,3i} = 2\omega_{2i} C_{2i} C_{3i}. \quad (67)$$

The network sink reactions use ω_i for every interstitial species, and the small clusters dissolve fully to monomers,

$$R_{v,s} = \omega_v C_v C_s^v, \quad R_{i,s} = \omega_i C_i C_s^i, \quad \mathcal{E}_{2i} = \omega_{2i} e_{2i} C_{2i}, \quad (68)$$

$$R_{2i,s} = \omega_i C_{2i} C_s^i, \quad R_{3i,s} = \omega_i C_{3i} C_s^i, \quad \mathcal{E}_{3i} = \omega_{3i} e_{3i} C_{3i}. \quad (69)$$

The displacement rate G (dpa s⁻¹) is partitioned among free monomers, small clusters and cascade-born loops so that the interstitial and vacancy atom budgets each close to G ,

$$G_v = G(1 - \varepsilon_{vL}), \quad G_i = G(1 - \varepsilon_{2i} - \varepsilon_{3i} - \varepsilon_{iL}), \quad (70)$$

$$G_{2i} = G\varepsilon_{2i}, \quad G_{3i} = G\varepsilon_{3i}, \quad (71)$$

$$G_{iL} = G\varepsilon_{iL}(1 - f_a), \quad G_{aiL} = G\varepsilon_{iL}f_a, \quad (72)$$

$$G_{vL} = G\varepsilon_{vL}(1 - f_a), \quad G_{avL} = G\varepsilon_{vL}f_a, \quad (73)$$

where $\varepsilon_{2i}, \varepsilon_{3i}$ are the di- and tri-interstitial cascade fractions, ε_{iL} the cascade a -loop fraction, ε_{vL} the cascade c -loop fraction, and f_a the aligned fraction. In the monomer-cluster equations the di- and tri-interstitial sources enter *per cluster*, as $G_{2i}/2$ and $G_{3i}/3$, and the atom balances satisfy $G_i + G_{2i} + G_{3i} + G_{iL} + G_{aiL} = G$ and $G_v + G_{vL} + G_{avL} = G$. An applied normal stress σ_n aligns a fraction of the loops,

$$f = \frac{\exp(n\sigma_n\Omega/k_BT) - 1}{\exp(n\sigma_n\Omega/k_BT) + 2}, \quad n \approx 10, \quad f_a = \frac{1}{3}(1 + 2f), \quad f_{na} = 1 - f_a, \quad (74)$$

which is the orientation dependence that underlies irradiation growth and creep.

Each loop grows by absorbing its own defect type and shrinks by absorbing the opposite type, which annihilates a stored defect. With $\ell \equiv z_c\Omega/(2\pi a^2)$, $\ell_a = \sqrt{\Omega/\pi b^a}$ and $\ell_c = \sqrt{\Omega/\pi b^c}$, the absorption prefactors are

$$p_{iL} = \frac{\ell_c}{\ell} \sqrt{c_{iL}^i n_{iL}}, \quad p_{aiL} = \frac{\ell_c}{\ell} \sqrt{c_{aiL}^i n_{aiL}}, \quad (75)$$

$$p_{vL} = \frac{\ell_c}{\ell} Q \sqrt{n_{vL} c_{vL}^v}, \quad p_{avL} = \frac{\ell_c}{\ell} Q \sqrt{n_{avL} c_{avL}^v}, \quad (76)$$

in which the *same* ℓ_c/ℓ scale is applied to all four families and Q is the c -loop absorption efficiency. The $\sqrt{cn}$ dependence is the sink strength $2\pi rN$ rewritten in terms of density and content, since $r \propto \sqrt{n}$

with $m = c/n$ the mean number of defects per loop. The species-resolved absorption components are then

$$iL_i = p_{iL} Z_i^a \phi_i, \quad iL_v = p_{iL} Z_v^a \phi_v g(r_{iL}), \quad (77)$$

$$aiL_i = p_{aiL} Z_i^a \phi_i, \quad aiL_v = p_{aiL} Z_v^a \phi_v g(r_{aiL}), \quad (78)$$

$$vL_v = p_{vL} Z_v^c \phi_v, \quad vL_i = p_{vL} Z_i^c \phi_i, \quad (79)$$

$$avL_v = p_{avL} Z_v^c \phi_v, \quad avL_i = p_{avL} Z_i^c \phi_i, \quad (80)$$

and the net loop-content growth rates and free-pool debits are

$$\dot{G}_{iL}^c = iL_i - iL_v, \quad \dot{G}_{aiL}^c = aiL_i - aiL_v, \quad (81)$$

$$\dot{G}_{vL}^c = vL_v - vL_i, \quad \dot{G}_{avL}^c = avL_v - avL_i, \quad (82)$$

$$\text{loop_abs}_i = iL_i + aiL_i + vL_i + avL_i, \quad \text{loop_abs}_v = vL_v + avL_v + iL_v + aiL_v, \quad (83)$$

$$\text{loop_recomb} = iL_v + aiL_v + vL_i + avL_i. \quad (84)$$

At high temperature and dose the vacancy flux can exceed the interstitial flux on a -loops, flipping their net climb negative and collapsing the reconstructed radius to zero. A C^1 -continuous smoothstep gate $g(r) \in [0, 1]$ suppresses the a -loop vacancy-absorption channel as $r \rightarrow r_{\min}^a$,

$$g(r) = \begin{cases} 0, & x \leq 0, \\ x^2(3-2x), & 0 < x < 1, \\ 1, & x \geq 1, \end{cases} \quad x = \frac{r/r_{\min}^a - 1}{w_r}, \quad (85)$$

and it is applied to the whole vacancy-absorption term — in the growth rate, in loop_abs_v and in loop_recomb — so that the point-defect balance stays closed.

Interstitial loops nucleate from homogeneous clustering plus a cascade source, the interstitial atoms consumed by the clustering reactions being deposited as loop content so that loops are born at finite size:

$$\dot{N}_{iL}^{\text{nuc}} = f_{na}(R_{i,3i} + R_{2i,2i}) + \frac{G_{iL}}{n_{iL}^{\text{nuc}}}, \quad \dot{N}_{aiL}^{\text{nuc}} = f_a(R_{i,3i} + R_{2i,2i}) + \frac{G_{aiL}}{n_{iL}^{\text{nuc}}}, \quad (86)$$

$$\dot{N}_{vL}^{\text{nuc}} = \frac{G_{vL}}{n_{vL}^{\text{nuc}}}, \quad \dot{N}_{avL}^{\text{nuc}} = \frac{G_{avL}}{n_{vL}^{\text{nuc}}}, \quad (87)$$

$$\mathcal{N}_c = 4R_{i,3i} + 2R_{2i,2i} + 3R_{2i,3i}, \quad (\text{nucleation content}) \quad (88)$$

with n_{iL}^{nuc} and n_{vL}^{nuc} the interstitials or vacancies per nucleated cascade loop. Vacancy loops anneal with an Arrhenius lifetime $\tau_{vL}(T) = \tau_{vL}^0 \exp(E_a^{vL}/k_B T)$, and analogously τ_{avL} for the aligned family, whose parameters default to the non-aligned values; the dissolving embryos return their density and stored content to the free pool at the rates n_{vL}/τ_{vL} and c_{vL}^v/τ_{vL} , which are the same forms as the 3D annealing vectors $\mathbf{a}_n$ and $\mathbf{A}_c$ of Section 2. The original capture-volume factor $(1 - (V_{\text{cap}}/\Omega)n_{vL})$ on vacancy-loop nucleation is removed; vacancy-loop density is instead saturated by the temperature-dependent annealing term, and the cascade-lifetime estimate $\tau_{vL} \approx -r_{vL}(0)/\dot{r}_{vL}|_0$ is replaced by the Arrhenius law.

Beyond flux-driven growth, loops coalesce by two channels whose rate scales are set by the *absorbed-flux* climb speed, which, unlike the net growth velocity, stays positive at saturation:

$$v_{\text{abs}}^a = \max\left(\frac{\ell_a(\ell_c/\ell) Z_i^a \phi_i}{2}, 0\right), \quad v_{\text{abs}}^c = \max\left(\frac{\ell_c(\ell_c/\ell) Q Z_v^c \phi_v}{2}, 0\right). \quad (89)$$

For a population of number concentration n , content c , length scale ℓ_s and coefficients c_{LL}, c_{LN} , define $r = \ell_s \sqrt{c/n}$, $N = n/\Omega$, the loop-loop spacing $d_{LL} = (\Omega/n)^{1/3}$ and the network spacing $d_N = \rho_N^{-1/2}$. Capping the radius at the relevant spacing, $r_{LL} = \min(r, d_{LL})$ and $r_{LN} = \min(r, d_N)$, the Avrami overlap probabilities and rate scales are

$$\phi_{LL} = 1 - \exp(-\kappa_{LL} \frac{4}{3} \pi r_{LL}^3 N), \quad \phi_{LN} = 1 - \exp(-\kappa_{LN} \pi r_{LN}^2 \rho_N), \quad (90)$$

$$\nu_{LL} = c_{LL} v_{\text{abs}} N^{1/3}, \quad \nu_{LN} = c_{LN} v_{\text{abs}} \sqrt{\rho_N}, \quad (91)$$

giving the number and content loss rates

$$\Delta N_{LN} = \nu_{LN} \phi_{LN} n, \quad \Delta N = \nu_{LL} \phi_{LL} n + \Delta N_{LN}, \quad \Delta C = \nu_{LN} \phi_{LN} c. \quad (92)$$

The like-loop channel conserves content, two loops merging into a larger one, while the loop-network channel removes both number and content, booked as network sink absorption so that the point-defect balance stays closed. The a -loops use $v_{\text{abs}}^a, \ell_a, (c_{LL}^a, c_{LN}^a)$ and the c -loops $v_{\text{abs}}^c, \ell_c, (c_{LL}^c, c_{LN}^c)$.

The network density is itself a state variable, seeded at the grown-in, temperature-dependent material value, log-linear in T between anchors ρ_{300} and ρ_{600} ,

$$\log_{10} \rho_N(0) = \log_{10} \rho_{300} + (\log_{10} \rho_{600} - \log_{10} \rho_{300}) \frac{T - 300}{600 - 300}, \quad (93)$$

and it grows by the line length of loops absorbed into the network while recovering first-order toward that seed,

$$\frac{d\rho_N}{dt} = c_{\rho N} \frac{2\pi}{\Omega} \sum_{k \in \{iL, aiL, vL, avL\}} r_k \Delta N_{LN,k} - k_{\rho N}^{\text{rec}} (\rho_N - \rho_N(0)). \quad (94)$$

Collecting these ingredients, the implemented right-hand side for the twelve physical species is

$$\frac{dC_v}{dt} = G_v + \frac{c_{vL}^v}{\tau_{vL}} + \frac{c_{avL}^v}{\tau_{avL}} - (R_{i,v} + R_{2i,v} + R_{3i,v} + R_{v,s} + \text{loop_abs}_v), \quad (95)$$

$$\frac{dC_i}{dt} = G_i + R_{2i,v} + 2\mathcal{E}_{2i} + 3\mathcal{E}_{3i} - (R_{i,v} + R_{i,i} + R_{i,2i} + R_{i,3i} + R_{i,s} + \text{loop_abs}_i), \quad (96)$$

$$\frac{dC_{2i}}{dt} = \frac{1}{2}G_{2i} + \frac{1}{2}R_{i,i} + R_{3i,v} - (R_{2i,v} + R_{i,2i} + R_{2i,2i} + R_{2i,s}) - \mathcal{E}_{2i}, \quad (97)$$

$$\frac{dC_{3i}}{dt} = \frac{1}{3}G_{3i} + R_{i,2i} - (R_{3i,v} + R_{i,3i} + R_{3i,s} + R_{2i,3i}) - \mathcal{E}_{3i}, \quad (98)$$

$$\frac{dn_{iL}}{dt} = \dot{N}_{iL}^{\text{nuc}} - \Delta N_{iL}, \quad (99)$$

$$\frac{dn_{aiL}}{dt} = \dot{N}_{aiL}^{\text{nuc}} - \Delta N_{aiL}, \quad (100)$$

$$\frac{dn_{vL}}{dt} = \dot{N}_{vL}^{\text{nuc}} - \frac{n_{vL}}{\tau_{vL}} - \Delta N_{vL}, \quad (101)$$

$$\frac{dn_{avL}}{dt} = \dot{N}_{avL}^{\text{nuc}} - \frac{n_{avL}}{\tau_{avL}} - \Delta N_{avL}, \quad (102)$$

$$\frac{dc_{iL}^i}{dt} = \dot{C}_{iL}^c + f_{na} \mathcal{N}_c + G_{iL} - \Delta n_{iL}, \quad (103)$$

$$\frac{dc_{aiL}^i}{dt} = \dot{C}_{aiL}^c + f_a \mathcal{N}_c + G_{aiL} - \Delta n_{aiL}, \quad (104)$$

$$\frac{dc_{vL}^v}{dt} = \dot{C}_{vL}^c + G_{vL} - \frac{c_{vL}^v}{\tau_{vL}} - \Delta n_{vL}, \quad (105)$$

$$\frac{dc_{avL}^v}{dt} = \dot{C}_{avL}^c + G_{avL} - \frac{c_{avL}^v}{\tau_{avL}} - \Delta n_{avL}. \quad (106)$$

Once the concentrations are known the mean loop radii follow from the geometric content relations $r_{XL} = \ell^X \sqrt{c_{XL}^x / n_{XL}}$ with $\ell^X = \ell_a$ for the two interstitial families and ℓ_c for the two vacancy families, using the Zr length scales $\ell_a = 0.1525$ nm, $\ell_c = 0.0854$ nm, $\ell_a/\ell = 0.09$ and $\ell_c/\ell = 0.05$.

Six monotone integrals are carried alongside the physics, at state indices 12–17, purely as a diagnostic of atom conservation; they do not feed back. With the stored interstitial and vacancy atoms $I_{\text{st}} = C_i + 2C_{2i} + 3C_{3i} + c_{iL}^i + c_{aiL}^i$ and $V_{\text{st}} = C_v + c_{vL}^v + c_{avL}^v$,

$$\dot{\Sigma}_i^{\text{prod}} = G_i + 2G_{2i} + 3G_{3i} + G_{iL} + G_{aiL}, \quad \dot{\Sigma}_v^{\text{prod}} = G_v + G_{vL} + G_{avL}, \quad (107)$$

$$\dot{\Sigma}^{\text{rec}} = R_{i,v} + R_{2i,v} + R_{3i,v} + \text{loop_recomb}, \quad (108)$$

$$\dot{\Sigma}_i^{\text{sink}} = R_{i,s} + 2R_{2i,s} + 3R_{3i,s} + \Delta n_{iL} + \Delta n_{aiL}, \quad \dot{\Sigma}_v^{\text{sink}} = R_{v,s} + \Delta n_{vL} + \Delta n_{avL}, \quad (109)$$

and the residual between I_{st} (respectively V_{st}) and $\Sigma^{\text{prod}} - \Sigma^{\text{rec}} - \Sigma^{\text{sink}}$ quantifies the model's conservation error in post-processing.

The macroscopic observables follow from the loop contents. With strain factors A_a, A_c the growth and creep strains are

$$\varepsilon_a = A_a(c_{iL}^i + c_{aiL}^i), \quad \varepsilon_c = -A_c(c_{vL}^v + c_{avL}^v), \quad (110)$$

$$\dot{\varepsilon}_{\text{creep}} = (A_a c_{aiL}^i - \frac{1}{2} A_a c_{iL}^i) + (-A_c c_{avL}^v + \frac{1}{2} A_c c_{vL}^v), \quad (111)$$

and radiation hardening is evaluated with a dispersed-barrier model: for each obstacle population X of number density $N_X = C_X/\Omega$ and size d_X , $\Delta\tau_X = \alpha_X \mu b \sqrt{N_X d_X}$, the short-range contributions adding in quadrature as $\Delta\tau_{SR} = \sqrt{\sum_X \Delta\tau_X^2}$, the long-range network term being $\Delta\tau_{LR} = \alpha_{LR} \mu b \sqrt{\rho_N}$, and the yield increment $\Delta\sigma_y = (\Delta\tau_{SR} + \Delta\tau_{LR})/M$.

3.2 ZrMicro-3d

ZrMicro-3d carries four immobile families, each with a number density N_k (per b^3) and a stored defect content c_k (a volume fraction, numerically equal to the **ZrMicro-0d** per-atom content). The mean number of defects per loop, the loop radius and the sink strength follow as

$$m_k = \frac{c_k}{N_k \Omega}, \quad r_k = \sqrt{\frac{m_k \Omega}{\pi b |b_k|}}, \quad S_k = 2\pi r_k N_k, \quad (112)$$

with a sigmoidal interpolation between a bi-pyramid and a loop morphology below and above a critical size, as in Eq. (29). The evolution equations are

$$\frac{dN_k}{dt} = \underbrace{\frac{G \varepsilon_k}{n_k^{\text{nuc}} \Omega}}_{\text{cascade nucleation}} - \underbrace{(\nu_{LL} \phi_{LL} + \nu_{LN} \phi_{LN}) N_k}_{\text{coalescence}} - \underbrace{\frac{N_k}{\tau_{vL}} \delta_{k,v}}_{\text{dissolution}}, \quad (113)$$

$$\frac{dc_k}{dt} = \underbrace{G \varepsilon_k}_{\text{cascade seed}} + \underbrace{\dot{g}_k}_{\text{absorption}} - \underbrace{\nu_{LN} \phi_{LN} c_k}_{\text{loop-network}} - \underbrace{\frac{n_k^{\text{nuc}} N_k \Omega}{\tau_{vL}} \delta_{k,v}}_{\text{annealing release}} - \underbrace{\alpha_v(m_k) \delta_{k,v}}_{\text{emission}}, \quad (114)$$

where the absorption flux is

$$\dot{g}_k = \sum_m \bar{D}_m Z_{km} S_k c_m |m_m| \text{sgn}_k(m) \quad (115)$$

with a smooth gate suppressing the shrinking channel as $r_k \rightarrow r_{\min}$, and the Avrami overlap factors are those of the reduced model,

$$\phi_{LL} = 1 - e^{-\kappa_{LL} \frac{4}{3} \pi r_{LL}^3 N_k}, \quad \phi_{LN} = 1 - e^{-\kappa_{LN} \pi r_{LN}^2 \rho_N}, \quad (116)$$

the like-loop channel again conserving content and removing only density while the loop-network channel removes both. The DAD capture efficiencies are generated from a single anisotropy parameter per mobile species,

$$Z_m^{(c)} = Z_m^0 p_m, \quad Z_m^{(a)} = \frac{Z_m^0}{2} (p_m + p_m^{-2}), \quad (117)$$

with p_m and Z_m^0 chosen so that Eq. (117) reproduces the empirical symmetric forms of Eq. (60) used by **ZrMicro-0d**.

One term in Eq. (114) departs from the obvious form and is worth isolating, because it controls the mean $\langle c \rangle$ loop size. The dissolution lifetime τ_{vL} describes the disappearance of still-small *embryo* loops, loops that survive and grow being stable, so each dissolving loop returns its *birth* content of n^{nuc} defects rather than its current mean content, and the release is proportional to the density rather than to the content. The consistency of that choice is easily checked: at number saturation $N^* = \dot{N}_{\text{nuc}} \tau_{vL}$, so the release becomes $n^{\text{nuc}} N^* \Omega / \tau_{vL} = n^{\text{nuc}} \dot{N}_{\text{nuc}} \Omega = G \varepsilon_k$, which cancels the cascade content seed exactly and leaves the net content growth purely flux-driven. Had the release been written as c_k / τ_{vL} the mean loop size would be pinned near n^{nuc} ; in practice that gave $m_{\langle c \rangle} \approx 490$ against $n^{\text{nuc}} = 400$, where **ZrMicro-0d** predicts $m_{\langle c \rangle} \approx 1.7 \times 10^5$ — a discrepancy of more than two orders of magnitude in the stored $\langle c \rangle$ content, and hence in the c -axis growth strain.

Because the two codes use different primitive variables, every binary reaction coefficient must be mapped rather than transcribed. MoDELib2-NNL builds its coefficients as

$$K_{ab} = p_{ab} \frac{4\pi(r_a + r_b)(D_a + D_b)}{\Omega}, \quad (118)$$

whereas ZrMicro-0d uses $K_{ab} = f_{ab}(\omega_a + \omega_b)$. Equating the two and using Eq. (3) gives the prefactors that must be supplied to the 3D code,

$$p_{ab} = f_{ab} \frac{z_c}{a^2} \frac{\Omega}{4\pi(r_a + r_b)} \frac{D_a^{0D} + D_b^{0D}}{D_a^{3D} + D_b^{3D}} \quad (119)$$

in which the final ratio is unity except for the di- and tri-interstitials, which transport at the monomer diffusivity for the reason given below. The mapping is directly verifiable, because MoDELib2-NNL prints its assembled interaction matrices in s^{-1} at start-up; Table 3 compares them with the ZrMicro-0d expressions.

Table 3: Assembled second-order interaction coefficients of ZrMicro-3d, compared with the ZrMicro-0d expressions at 573 K.

entry	ZrMicro-3d [s^{-1}]	ZrMicro-0d [s^{-1}]	difference
(v, i)	2.526876×10^7	$\frac{1}{2}(\omega_i + \omega_v) = 2.52560 \times 10^7$	0.05 %
(i, i)	-2.021161×10^8	$-4\omega_i = -2.020216 \times 10^8$	0.05 %
$(v, 2i)$	6.940722×10^3	$\omega_{2i} + \omega_v = 6.93473 \times 10^3$	0.09 %

The ZrMicro-3d branch modifies MoDELib2-NNL in three ways, leaving Po’s Zr4.txt untouched and placing the reconciled definition in Zr3d_ghoniem.txt. The first is physics that was absent altogether. Raising iSize from 0 to 8, by way of the intermediate iSize = 4 stage carrying contents alone, turns on four immobile families with an evolving density and a stored content each, and solveImmobileClusters(), previously an empty stub, now implements Eqs. (113)–(116). A new evaluator supplies the loop-absorption sink marked // Missing immobile sinks in the original source, excluding the grain boundary because the Dirichlet condition imposes it spatially, and loopDADbias() implements Eq. (117). The finite-element node coordinates are also written once per run, without which the fields cannot be located in space: the cluster-dynamics trial functions live on second-order elements, so the field rows do not correspond to the mesh vertices.

The second is a set of corrections, of which the most consequential is an uninitialized read. The routine assembling the first-order reaction matrix reads the binding-energy array, which was declared *after* the matrix in the class; since C++ initializes members in declaration order, the Boltzmann factor $\exp(-E_b/k_B T)$ evaluated to unity instead of 3.8×10^{-9} , inflating the dissociation rate to the bare attempt frequency and suppressing C_{2i} and C_{3i} by 2.4×10^6 . It is a defect of the original code, invisible in the distributed calibration because every binding energy was set to 5 eV, and it should be reported upstream. Two corrections concern the atomic volume: the polycrystal class returns $\det(\text{lattice basis})$, the volume of a lattice *cell*, which in hexagonal Zr holds two atoms, so every content–radius–density conversion inherited the error; and the flux, formed as $-D/\Omega$ and multiplied back by an atomic volume in the weak form, acquired a spurious factor of 3.98 relative to every reaction term once those two ceased to be the same quantity. Both now use one Ω , supplied by an optional material-file key. Vacancy-loop dissolution and Avrami coalescence were integrated explicitly, violating the stability limit at the dose step used ($\Delta t/\tau_{vL} \approx 35$, $\nu_{LN}\phi_{LN}\Delta t \approx 3.6$) so that each sub-step overshoot and clamped to the positivity floor; both are first-order in their own variables and are now implicit. The immobile Burgers vectors are lattice components, where Cartesian ones had been supplied, giving $|b| = 1.0, 0.753, 1.197$ across the three prismatic families instead of unity — caught by the zero-stress symmetry check, which is available only because those variants are equivalent without stress. Finally, the minimum stable radius was compared in metres against radii in units of b , leaving the shrinking gate permanently open, and the loop density was accumulated per atom where the geometric helpers expect it per b^3 , inflating the mean loop size by 2.82.

The third is parameter reconciliation: Table 4 lists the material-file entries that differ from the distributed MoDELib2-NNL values, every original preserved in a comment in the file.

Table 4: Material parameters changed to match ZrMicro-0d.

quantity	MoDELlib2-NNL	ZrMicro-0d	basis
surviving efficiency	0.01	1.0	ZrMicro-0d loses defects only to in-cascade clustering
cascade fractions	1, 0.6, 0.3212, 0.0788	0.995, 0.9779, 0.005, 0.00333	$G_v/G = 1 - \varepsilon_{vL}$, etc.
E^m (SIA)	0.09 eV	0.759101 eV	ZrMicro-0d fitted value
E^m (vacancy)	0.9 eV	1.2 eV	ZrMicro-0d
D_0	$4.96/8.19 \times 10^{-10}$	5.21645×10^{-7}	$D_0 = a^2 \nu$
E_v^f	1.5 eV	1.6 eV	gives $C_v^{\text{eq}} = 8.45 \times 10^{-15}$
Ω	$4.78 \times 10^{-29} \text{ m}^3$	$1.2 \times 10^{-29} \text{ m}^3$	ZrMicro-0d
reaction prefactors	2.856, 1, 0	Eq. (119)	Table 3
E_b (2i, 3i)	5 eV	0.957, 2.0 eV	ZrMicro-0d
network sink	0	$\rho_N, Z_N \rho_N$	ZrMicro-0d Eq. (62)

Two of those are modelling choices rather than transcriptions. The di- and tri-interstitials transport at the *monomer* diffusivity rather than at the value implied by $E_{2i}^m = 1.36 \text{ eV}$, following ZrMicro-0d, which is inconsistent on the point but unambiguous about where each frequency is used: clusters take ω_i in the network-sink and loop-absorption terms, while ω_{2i} enters only the cluster-cluster reactions, whose rates the last factor of Eq. (119) recovers exactly. The alternative makes $D_{2i} \sim 5 \times 10^{-6} D_i$ and, since Eq. (1) carries no mass term, leaves the cluster block of the Newton matrix singular. The second is the loop sink convention: ZrMicro-0d applies the $\langle c \rangle$ prefactor ℓ_c to all four families and a fitted factor Q on the $\langle c \rangle$ channel, whereas ZrMicro-3d builds r_k from each family's own $|b_k|$, so per-family multipliers of 0.2915 and 0.7923 restore the ZrMicro-0d convention exactly, applied identically to the sink and to the growth flux it causes.

3.3 Correspondence between the codes

The two models describe the same physics at two levels of spatial resolution, and the correspondence between them is an identification of the 0D scalars with volume averages of the 3D fields,

$$C_x \longleftrightarrow \langle c^x \rangle = \frac{1}{V} \int_V c^x(\mathbf{x}) dV, \quad x \in \{v, i, 2i, 3i\}, \quad (120)$$

under which the two share the immobile loop families, the loop number densities and the per-loop contents, together with the evolving network density ρ_N . The typeface convention of Section 2 has no analogue in the 0D model, where every quantity is a scalar, so the correspondence maps each 3D array onto the collection of its 0D components.

The central structural difference is transport. The 3D master equation (1) carries the divergence $-\nabla \cdot \mathbf{J}$ explicitly, whereas the 0D model integrates the transient system $\dot{\mathbf{y}} = \mathbf{f}(\mathbf{y}; \mathbf{p}, T)$ with no transport at all. The two reduce to one another when the divergence is replaced by its volume average: for a fixed representative volume the divergence integrates to the net boundary flux, so

$$-\frac{1}{V} \int_V \nabla \cdot \mathbf{J}^x dV \longleftrightarrow -R_{x,s} - \text{loop_abs}_x, \quad (121)$$

that is, the spatial sink term of the 3D PDE is represented in 0D by the lumped network-sink and loop-absorption rates. With this replacement, and reading the 0D system at steady state, the component-wise expansion of Eq. (1) reproduces the 0D right-hand side for each of the twelve physical species term by term.

Table 5 collects the mapping. Five of the operators are strictly identical once Eq. (120) is granted — cascade production, the second-order reactions, thermal emission, the network sinks and the point-defect fluxes — and the individual identities were established alongside each operator in Section 2. Four require interpretation, and they are the terms through which the two models genuinely differ.

The first is loop absorption. In 3D it is expressed as the sink strengths k_m^2 of Eq. (19), built from the bias factors and family radii; in 0D it is the explicit mass-conserving coupling of Eq. (76). The two are constructed to debit the mobile species by exactly the amount credited to the loop content, so they are physically equivalent although algebraically distinct. The second is the DAD bias, which

is the same physical effect parameterized in two ways: the 3D model derives it from the anisotropy ratio $p^m = (D_c^m/D_a^m)^{1/6}$ through Eq. (20), obtaining the bias from the tensorial flux itself, whereas the 0D model introduces the phenomenological parameters δ_i, δ_v of Eq. (60). The two coincide in the isotropic limit ($p^m \rightarrow 1, \delta \rightarrow 0$), where both recover the unbiased capture efficiency; away from it the mapping $\delta \leftrightarrow (p, Z^0)$ is a modelling choice and must be interpreted rather than taken as an identity.

The remaining two are closures rather than reparameterizations. The mobile-species boundary condition (16), with the stress-dependent equilibrium concentration of Eq. (18), encodes the spatially-resolved coupling to surfaces and internal interfaces; in 0D this is subsumed into the lumped network sink and the spatially uniform equilibrium concentration $C_v^e = \exp(-E_v^f/k_B T)$. And the loop radii, which in 3D follow from the resolved geometry, are recovered in 0D through the geometric content closure $r_{X\mathcal{L}} = \ell \sqrt{c_{X\mathcal{L}}^x/n_{X\mathcal{L}}}$. These two are exactly the point at which the reduction from three dimensions to zero introduces additional modelling assumptions, and they are the natural place to test the fidelity of the reduced model against the resolved one.

Table 5: Correspondence between the 3D spatially-resolved model (Section 2) and the 0D reduced model (Section 3.1).

Quantity / term	ZrMicro-3d	ZrMicro-0d	Relation
Mobile concentrations	$\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$ (fields)	C_v, C_i, C_{2i}, C_{3i} (scalars)	volume average
Transport	$-\nabla \cdot \mathbf{J}, \mathbf{J}^x = -D^x \nabla c^x$	lumped $-R_{x,s} - \text{loop_abs}_x$	interpretation
Production	$\mathbf{P}_M = G\epsilon \mathbf{f}$	$\{G_v, G_i, \frac{1}{2}G_{2i}, \frac{1}{3}G_{3i}\}$	identical
Second-order reactions	$\frac{1}{2}\mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$	$R_{a,b} = (\omega_a + \omega_b)C_a C_b$	identical
Thermal emission	$\mathbf{E} \mathbf{c}_M$	$\mathcal{E}_{2i}, \mathcal{E}_{3i}$	identical
Network sinks	$\text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\}$	$R_{x,s} = \omega_x C_x C_s^x$	identical
Point-defect fluxes	ϕ_v, ϕ_i	ϕ_v, ϕ_i	identical
Loop absorption	$\mathbf{D}_d \mathbf{K}_{\mathcal{L}}, k_m^2$	loop_abs_x (mass-conserving)	equivalent (interpretation)
DAD loop bias	$p^m = (D_c^m/D_a^m)^{1/6}, Z^{\text{DAD},0}$	δ_i, δ_v	interpretation
Boundary condition	Dirichlet $c_{\text{bnd,eq}}^k$, Eq. (16)	lumped sink, C_v^e	closure
Loop radii	resolved geometry	$r_{X\mathcal{L}} = \ell \sqrt{c_{X\mathcal{L}}^x/n_{X\mathcal{L}}}$	closure

4 Parameter Identification for the ZrMicro-0d CD Model

The **ZrMicro** model evolves a vector of point-defect and defect-cluster concentrations under irradiation. A set of physically meaningful but imperfectly known parameters $\mathbf{p} \in \mathbb{R}^p$ (DAD bias factors, migration and binding energies, attempt frequencies, cascade efficiencies, loop-annealing lifetimes, coalescence coefficients, dose rate, network-density anchors, ...) controls the kinetics. We are given a database of M experimental measurements obtained at various irradiation temperature/dose combinations (T_m, dpa_m), each reporting a total dislocation-loop number density N_L^m (m^{-3}) and/or a mean loop diameter d^m (nm). The task is the classical *inverse problem*: find the parameter vector that minimizes the discrepancy between simulated and measured observables across the whole database, subject to physical bounds

$$\mathbf{p}^* = \arg \min_{\mathbf{p} \in [\mathbf{p}_{\min}, \mathbf{p}_{\max}]} J(\mathbf{p}), \quad (122)$$

where J is the misfit functional defined in §4.2 and the box $[\mathbf{p}_{\min}, \mathbf{p}_{\max}]$ is taken from the **Params** worksheet.

Equation (122) is a derivative-free, multimodal, bound-constrained global optimization problem: the forward map is a stiff nonlinear ODE solver and is moderately expensive (~ 7 s per objective evaluation), so sample efficiency and robustness to local minima both matter.

4.1 Forward model, observation operator, and identification targets

We begin by laying out the ingredients of the inverse problem. The reduced cluster-dynamics model of the previous section acts as the forward map: given a parameter vector it integrates the rate-theory state in time and returns the predicted microstructural evolution. An observation operator then projects this internal state onto the quantities that are actually measured, so that simulations and experiments can be compared on the same footing. Finally, the input workbook collects the target database and the list of parameters to be identified, defining the data against which the model is calibrated. The forward map is the rate-theory model of Section 3.1, integrated in time. Its state is augmented beyond the 12 physical concentrations: it also carries six monotone point-defect conservation accumulators and the evolving network dislocation density, for a total of 19 ODEs,

$$\mathbf{y}(t) = \left[\underbrace{C_v, C_i, C_{2i}, C_{3i}, C_{iL}, C_{aiL}, C_{vL}, C_{avL}, C_{iL}^i, C_{aiL}^i, C_{vL}^v, C_{avL}^v}_{12 \text{ physical}}, \underbrace{\Sigma_i^{\text{prod}}, \dots, \Sigma_v^{\text{sink}}}_{6 \text{ accumulators}}, \rho_N \right]^\top \in \mathbb{R}^{19}, \quad (123)$$

collecting free vacancy/interstitial concentrations, small interstitial clusters, the number densities of interstitial and vacancy loops (pure and aligned), the point-defect content of those loops (all in atomic-fraction units), the conservation diagnostics, and the network density (m^{-2}). Only the 12 physical components enter the observation operator. The evolution obeys the coupled rate equations of Appendix A,

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(\mathbf{y}; \mathbf{p}, T), \quad \mathbf{y}(0) = \mathbf{y}_0(\mathbf{p}, T), \quad (124)$$

where $\mathbf{f}$ encodes generation by displacement cascades, mutual recombination, biased (DAD) absorption at sinks and loops, cluster nucleation, thermal emission, loop annealing, and geometric loop coalescence. Temperature enters through the Arrhenius jump frequencies, the equilibrium concentration, the DAD bias, the loop lifetimes, and the seed network density,

$$\omega_i = z_c \nu_i e^{-E_i^m/k_B T}, \quad \omega_v = z_c \nu_v e^{-E_v^m/k_B T}, \quad C_v^{\text{eq}} = e^{-E_v^f/k_B T}, \quad (125)$$

so a change of T requires recomputing all derived quantities and re-integrating (124). The system is solved with a stiff integrator (SUNDIALS CVODE/BDF, or SciPy LSODA for the build-free path) to obtain $\mathbf{y}(t)$ on a logarithmically spaced time grid $\{t_k\}_{k=1}^{N_t}$.

Irradiation dose accumulates linearly with the dose rate G (a member of $\mathbf{p}$),

$$\text{dpa}(t) = G t \quad [\text{dpa}], \quad (126)$$

which maps the time axis onto a dose axis. To compare against a measurement taken at dose dpa_m , we evaluate the model at the matching time $t_m = \text{dpa}_m/G$. In practice the integration horizon for a given temperature group is chosen so that the maximum requested dose is comfortably covered, $t_{\text{end}} = 10^{\lceil \log_{10}(3 \text{ dpa}_{\text{max}}/G) \rceil}$, and the observables are read by interpolation in $\log_{10} \text{dpa}$ (see below).

What an experiment measures is not that state but a projection of it, and the observation operator supplies the projection. The observables are now recovered *separately for the two loop families*, matching the split experimental database: the *a*-type (interstitial, prismatic) loops are compared against **Targets_A** and the *c*-type (vacancy, basal) loops against **Targets_C**. Within each family the pure and aligned variants are first *combined*, then reduced to a density and a number-weighted mean diameter. Writing $k \in \{A, C\}$ for the loop family, the operator $\mathbf{h}^k = (h_{N_L}^k, h_d^k)$ is

$$h_{N_L}^A(\mathbf{y}) = \frac{1}{\Omega} (C_{iL} + C_{aiL}), \quad h_{N_L}^C(\mathbf{y}) = \frac{1}{\Omega} (C_{vL} + C_{avL}), \quad [\text{m}^{-3}], \quad (127)$$

$$h_d^A(\mathbf{y}) = 2 \frac{C_{iL} r_{iL} + C_{aiL} r_{aiL}}{C_{iL} + C_{aiL}} \times 10^9, \quad h_d^C(\mathbf{y}) = 2 \frac{C_{vL} r_{vL} + C_{avL} r_{avL}}{C_{vL} + C_{avL}} \times 10^9, \quad [\text{nm}], \quad (128)$$

where Ω is the atomic volume and the loop radii follow from the loop content via the geometric relations $r_{iL} = \ell_a \sqrt{C_{iL}^i/C_{iL}}$, $r_{vL} = \ell_c \sqrt{C_{vL}^v/C_{vL}}$ (and analogously for the aligned variants r_{aiL}, r_{avL}), with ℓ_a, ℓ_c the loop length scales. Equation (127) gives the per-family loop density (interstitial loops for *A*, vacancy loops for *C*); (128) is the per-family, number-weighted mean diameter, combining the pure and aligned sub-populations. The two families are *not* merged into a single total — they are fit against their own target sheets and only later combined in the weighted objective $J = w_A J_A + w_C J_C$ (§4.2).

Collecting the forward solve, the dose mapping, and the observation operator, the *predicted* observables at condition m — which belongs to a definite loop family $k(m)$ given by its source sheet — are written compactly as

$$(\widehat{N}_L^m, \widehat{d}^m) = \mathcal{G}_m(\mathbf{p}) \equiv \mathbf{h}^{k(m)}(\mathbf{y}(t_m = \text{dpa}_m/G; \mathbf{p}, T_m)). \quad (129)$$

Because $\mathcal{G}_m$ depends on temperature only through T_m , all measurements sharing a temperature (across *both* families) are served by a *single* ODE integration, with (129) obtained by interpolation at the several doses of that group.

It remains to say where the targets and the parameter list come from. The identification is driven entirely by the extended workbook `Zr_input_parameters.xlsx`. Two families of experimental loops are fit *simultaneously* but kept as distinct target sets:

- **Targets_A** — *a*-type (prismatic, interstitial) loops: 47 usable points over 9 temperatures (473–673 K), 45 reporting a density and 35 a diameter, compiled from Northwood, Jostsons, Gilbert, Harte and Kobylansky. The high-temperature ($T \geq 650$ K) *diameter* rows are excluded (set to NaN) because those large “*a*” loops are vacancy-character and would otherwise contaminate the interstitial-loop size fit.
- **Targets_C** — *c*-type (basal, vacancy) loops: 5 points over 2 temperatures (573, 583 K), all reporting both density and diameter (Harte 2017, Tournadre 2012).

The **Params** worksheet *is* the optimization list: every row carrying a valid $[\theta_{\min}, \theta_{\max}]$ pair becomes a free parameter; any other quantity is held fixed at its workbook value. The present list has $p = 23$ free parameters (Table 6). A parameter is searched in $\log_{10}$ when its range ratio $\theta_{\max}/\theta_{\min} \geq \kappa$ with $\kappa = 20$ (LOG_RATIO_THRESH); otherwise it is searched linearly (Eq. (136)).

4.2 Objective functional and parameter normalization

This subsection specifies how the mismatch between model predictions and targets is measured and how the parameters are conditioned for optimization. We first construct the scalar objective functional that aggregates the weighted residuals across all observed quantities into a single quantity to be minimized, and then introduce the transformation and normalization of the parameters that maps physically disparate quantities onto a common, well-scaled search domain so that the optimizer treats every degree of freedom on an equal footing. The implemented objective is a *per-(loop-type, temperature)-group* misfit, not a flat sum over measurements — this prevents the dense low/mid- T *a*-loop cluster from swamping the sparse high- T and *c*-loop data. Index loop types by $k \in \{A, C\}$ and let $\mathcal{T}_k$ be the temperatures present in set k . For a group (k, T) with experimental points $\{(\text{dpa}_m, N_L^m, d^m)\}$, the model is read at the measured doses (interpolating in $\log_{10}$ dpa) and the density and diameter mean-squared errors are

$$\text{MSE}_{k,T}^N = \frac{1}{|\mathcal{M}_{k,T}^N|} \sum_{m \in \mathcal{M}_{k,T}^N} (\log_{10} \widehat{N}_L^m - \log_{10} N_L^m)^2, \quad (130)$$

$$\text{MSE}_{k,T}^d = \frac{1}{|\mathcal{M}_{k,T}^d|} \sum_{m \in \mathcal{M}_{k,T}^d} \left(\frac{\widehat{d}^m - d^m}{d^m} \right)^2, \quad (131)$$

(log metric for density, relative metric for diameter). The group score averages whichever of the two are present,

$$s_{k,T} = \text{mean} \{ w_N \text{MSE}_{k,T}^N, w_d \text{MSE}_{k,T}^d \}, \quad (132)$$

and each group carries weight $w_{k,T} = w_k n_T$ with loop-type weights $w_A = w_C = 1$ and a high-temperature boost $n_T = 2$ for $T \geq 650$ K (else 1). The data misfit is the weight-normalised group average (“balance by temperature”),

$$J_{\text{data}}(\mathbf{p}) = \frac{\sum_k \sum_{T \in \mathcal{T}_k} w_{k,T} s_{k,T}}{\sum_k \sum_{T \in \mathcal{T}_k} w_{k,T}} \equiv w_A J_A + w_C J_C. \quad (133)$$

A *smooth-rise (anti-overshoot) penalty* is added to reward monotonic saturation and punish peak-then-collapse trajectories. For each loop density/diameter history $v(\text{dpa})$ (with $v = \log_{10} N_L$ for densities, $v = d$ for diameters), restricted to $\text{dpa} \geq \text{dpa}_{\text{lo}} = 10^{-3} \text{ dpa}$, the relative downward total variation is

$$\mathcal{O}[v] = \frac{-\sum_{j: \Delta v_j < 0} \Delta v_j}{\max(v) - \min(v)} \in [0, 1], \quad \Delta v_j = v_{j+1} - v_j, \quad (134)$$

(0 for a monotonic rise to a plateau, $\rightarrow 1$ for a fully retraced peak), and J_{os} is its mean over all loop trajectories. The total objective is

$$J(\mathbf{p}) = J_{\text{data}}(\mathbf{p}) + w_{\text{os}} J_{\text{os}}(\mathbf{p}) \quad (w_N = w_d = w_A = w_C = w_{\text{os}} = 1). \quad (135)$$

The density log-metric and diameter relative-metric are scale-invariant maximum-likelihood estimators for positive quantities spanning several decades [1]. If any forward solve in a parameter set fails to converge, the evaluation returns a large finite penalty $J = J_{\text{pen}}$ so the optimizer smoothly avoids that region.

The parameters at which that objective is evaluated need conditioning of their own, because several span orders of magnitude (e.g. $\rho \in [10^{12}, 10^{15}]$, $\tau_{vL} \in [10^3, 10^5]$, $\nu_v \in [2 \times 10^8, 10^{10}]$). To give the optimizer a well-conditioned, comparably scaled search space we apply a component-wise transform $\mathcal{T}$: a base-10 logarithm for parameters whose range ratio exceeds a threshold κ (default $\kappa = 20$), and the identity otherwise,

$$x_j = \mathcal{T}_j(\theta_j) = \begin{cases} \log_{10} \theta_j, & \theta_{j,\min} > 0 \text{ and } \theta_{j,\max}/\theta_{j,\min} \geq \kappa, \\ \theta_j, & \text{otherwise,} \end{cases} \quad \theta_j = \mathcal{T}_j^{-1}(x_j). \quad (136)$$

The optimizers operate on $\mathbf{x} = \mathcal{T}(\mathbf{p})$ within the transformed box $[\mathbf{x}_{\min}, \mathbf{x}_{\max}] = [\mathcal{T}(\mathbf{p}_{\min}), \mathcal{T}(\mathbf{p}_{\max})]$. For the surrogate solver we additionally normalize to the unit hypercube, $\mathbf{u} = (\mathbf{x} - \mathbf{x}_{\min}) \oslash (\mathbf{x}_{\max} - \mathbf{x}_{\min}) \in [0, 1]^p$, where $\oslash$ is the element-wise (Hadamard) division. Any proposal is projected back into the box before evaluation, $\mathbf{x} \leftarrow \min(\max(\mathbf{x}, \mathbf{x}_{\min}), \mathbf{x}_{\max})$.

4.3 Identification workflow and optimization algorithms

Having defined the forward model, the targets, and the objective, we now describe how the parameters are actually identified. The workflow couples a robust global exploration of the high-dimensional parameter space with an efficient local refinement, so that the procedure is neither trapped in poor local minima nor wasteful of expensive forward evaluations. Three complementary strategies are used: a population-based global search by differential evolution, a surrogate-assisted “self-learning” scheme that interpolates the objective with radial basis functions to cut the number of model calls, and a final Nelder–Mead simplex refinement of the best candidate. Figure 1 shows the overall pipeline; Figure 2 details the forward-evaluation block that is invoked for every candidate.

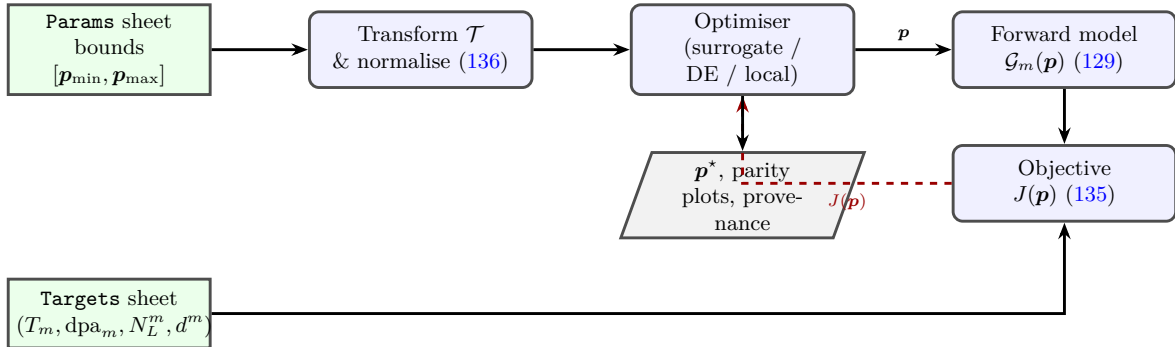


Figure 1: Parameter-identification workflow. The optimizer proposes a parameter vector, the forward model and observation operator predict the observables, the objective scores them against the experiments, and the (dashed) feedback drives the next proposal until convergence or budget exhaustion.

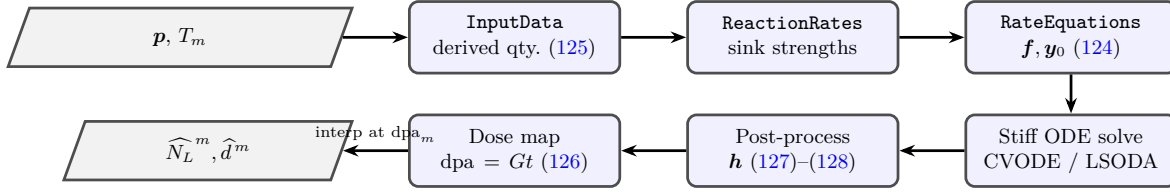


Figure 2: Forward evaluation $\mathcal{G}_m$ for one temperature group. One stiff integration produces the full time history; the observables at every measured dose of the group are then obtained by interpolation in $\log_{10}$ dpa.

Within that workflow the search itself is carried out in stages. Three complementary, SciPy-only [2] solvers are provided, selected by `OPT_CONFIG['method']`: an RBF *surrogate* self-learning loop (the default), *differential evolution*, and a purely *local* Nelder–Mead run. They share the same objective (135) and history log, and a final local Nelder–Mead polish (≤ 100 iterations) refines whichever global search is used. Each objective evaluation solves *all* target temperatures of one parameter set in a single OpenMP-parallel C++ (SUNDIALS CVODE/BDF) batch subprocess ($n_t=100$, $r_{\text{tol}}=10^{-6}$, $a_{\text{tol}}=10^{-20}$), giving roughly a $6\times$ speed-up over one subprocess per temperature.

The global stage is the first of these. Differential Evolution (DE) [3] is a robust, derivative-free, population-based global optimizer. A population $\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(NP)}\} \subset [\mathbf{x}_{\min}, \mathbf{x}_{\max}]$ is initialized by a low-discrepancy (Sobol’) sequence. At each generation, every target vector $\mathbf{x}^{(s)}$ produces a mutant by combining distinct random members.

$$\mathbf{v}^{(s)} = \mathbf{x}^{(r_1)} + F(\mathbf{x}^{(r_2)} - \mathbf{x}^{(r_3)}), \quad r_1 \neq r_2 \neq r_3 \neq s, \quad (137)$$

with differential weight $F \in (0, 2)$. Binomial crossover mixes the mutant and the target.

$$u_j^{(s)} = \begin{cases} v_j^{(s)}, & \text{if } \text{rand}_j \leq C_r \text{ or } j = j_{\text{rand}}, \\ x_j^{(s)}, & \text{otherwise,} \end{cases} \quad (138)$$

With a crossover probability C_r , greedy selection retains the better of the trial and target.

$$\mathbf{x}^{(s)} \leftarrow \begin{cases} \mathbf{u}^{(s)}, & J(\mathbf{u}^{(s)}) \leq J(\mathbf{x}^{(s)}), \\ \mathbf{x}^{(s)}, & \text{otherwise.} \end{cases} \quad (139)$$

DE explores the box thoroughly and is insensitive to local minima at the cost of many objective evaluations ($\mathcal{O}(NP \times \text{generations})$); the code defaults to a population factor of 6 and 15 generations, Sobol’-initialised.

The second stage replaces brute-force sampling by a learned model of the objective. When each evaluation is expensive, surrogate-based optimization (SBO) [4, 5] is far more sample-efficient: a cheap regression model is *learned* from the evaluated points and minimized to suggest where to sample next, so the budget is spent where it most reduces the objective. We use a radial-basis function (RBF) interpolation [6, 7, 8], which is the role a trained neural network would play in a learned-surrogate setting.

The loop opens with an initial design. An n_0 -point Latin-Hypercube design [9] $\{\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(n_0)}\}$ (with the nominal point injected) is evaluated, producing data $\mathcal{D} = \{(\mathbf{u}^{(k)}, y^{(k)})\}$, $y^{(k)} = J(\mathcal{T}^{-1}(\mathbf{x}^{(k)}))$. By default $n_0 = 2p + 1$.

Given $\mathcal{D}$, the RBF model that is fitted to it reads

$$s(\mathbf{u}) = \sum_{k=1}^n \lambda_k \psi(\|\mathbf{u} - \mathbf{u}^{(k)}\|) + q(\mathbf{u}), \quad \psi(r) = r^2 \log r \quad (\text{thin-plate spline}), \quad (140)$$

where q is a low-order polynomial tail. The coefficients $(\boldsymbol{\lambda}, q)$ solve the (regularized) interpolation system

$$\begin{bmatrix} \boldsymbol{\Psi} + \sigma \mathbf{I} & \mathbf{P} \\ \mathbf{P}^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ c \end{bmatrix} = \begin{bmatrix} \mathbf{y} \\ 0 \end{bmatrix}, \quad \Psi_{kl} = \psi(\|\mathbf{u}^{(k)} - \mathbf{u}^{(l)}\|), \quad (141)$$

With $\mathbf{P}$, the polynomial basis is evaluated at the samples with a small smoothing σ for numerical conditioning. The thin-plate spline is scale-free (no shape parameter to tune) and minimizes the second-derivative bending energy, giving a smooth response surface.

Each iteration thereafter minimizes the cheap surrogate over the unit cube to obtain a candidate.

$$\mathbf{u}_{\text{new}} = \arg \min_{\mathbf{u} \in [0,1]^p} s(\mathbf{u}), \quad (142)$$

solved with an inner DE on s (negligible cost). To prevent premature convergence, with a probability of p_e (default 0.25), the candidate is replaced by a uniformly random point — an explicit exploration/exploitation balance:

$$\mathbf{u}_{\text{new}} \leftarrow \begin{cases} \text{uniform } [0,1]^p, & \text{with prob. } p_e \text{ (explore),} \\ \arg \min_{\mathbf{u}} s(\mathbf{u}), & \text{with prob. } 1 - p_e \text{ (exploit).} \end{cases} \quad (143)$$

The new point is evaluated with the true model, appended to $\mathcal{D}$, and the surrogate is re-fit — the “self-learning” loop (Figure 3). The surrogate is a thin-plate-spline `RBFInterpolator` (smoothing 10^{-6}) and the infill minimisation (142) is itself an inner differential evolution on the cheap response surface. After $n_{\text{infill}} = 50$ infill iterations (default $n_0 = 2p + 1 = 47$ initial runs), the incumbent best is returned. The total true-model budget is $n_0 + n_{\text{infill}} \approx 97$ evaluations, typically an order of magnitude smaller than DE for comparable accuracy.

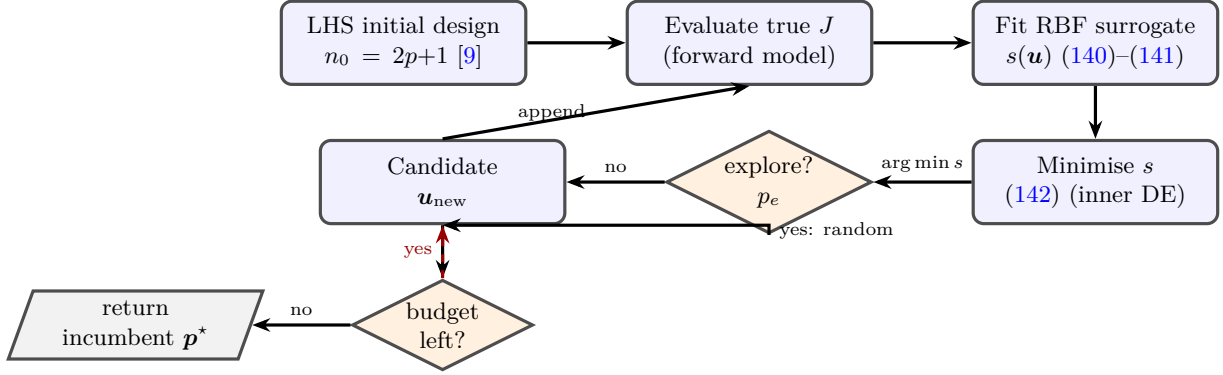


Figure 3: The RBF surrogate “self-learning” loop. The surrogate is re-trained on every newly evaluated point and minimised to choose the next sample, balancing exploitation (surrogate minimum) against exploration (random infill).

The RBF of (140) is only one choice for a learned surrogate. If `PyTorch/scikit-learn` is available, it may be replaced by a multilayer-perceptron or Gaussian-process regressor trained on $\mathcal{D}$; the acquisition (142) (optionally an expected-improvement criterion [4]) and the rest of the loop are unchanged. The RBF is chosen here for its zero external dependencies and robustness in moderate dimensions.

The third stage refines whatever the first two return. The incumbent from the global stage is polished with the derivative-free Nelder–Mead downhill-simplex method [10]. A simplex of $p + 1$ vertices is iteratively updated by reflection, expansion, contraction, and shrink operations about the centroid $\bar{\mathbf{x}}$ of its best p vertices,

$$\mathbf{x}_r = \bar{\mathbf{x}} + \alpha(\bar{\mathbf{x}} - \mathbf{x}_{p+1}) \quad (\text{reflection}), \quad (144)$$

$$\mathbf{x}_e = \bar{\mathbf{x}} + \gamma(\mathbf{x}_r - \bar{\mathbf{x}}) \quad (\text{expansion}), \quad (145)$$

$$\mathbf{x}_c = \bar{\mathbf{x}} + \beta(\mathbf{x}_{p+1} - \bar{\mathbf{x}}) \quad (\text{contraction}), \quad (146)$$

with standard coefficients $\alpha = 1$, $\gamma = 2$, $\beta = \frac{1}{2}$ (adaptive variant for higher p). Bounds are enforced by the projection in (136). Nelder–Mead converges quickly in the neighborhood of a minimum but is purely local—hence its use only as a final polish.

4.4 First-round optimization results

This subsection reports the outcome of the first complete identification round, in which the algorithms above are applied to the full target database. We first present the optimal parameter set returned by

the combined global–local search, then examine a reduced, minimal set of parameters that retains the identifiable degrees of freedom while suppressing those to which the objective is insensitive, and finally assess the overall quality of the fit against the experimental and reference targets.

Table 6 lists the full $p = 23$ free parameters with their search bounds, the $\log_{10}$ /linear search flag, and the optimum returned by the surrogate+Nelder–Mead run on the combined a - and c -loop database.

Table 6: Optimal parameter set after the first-round joint identification (surrogate + local polish). “log” marks parameters searched in $\log_{10}$ ($\theta_{\max}/\theta_{\min} \geq 20$). A bullet (•) in the last column flags a parameter that converged onto a bound (*railed*); these are unidentified by the present database and are dropped from the minimal set (Table 7).

Parameter	min	optimal	max	log	rail
δ_i^{300} (delta_DAD_i_300)	0.050	0.1533	0.450		
δ_i^{600} (delta_DAD_i_600)	0.020	0.0641	0.300		
Z_{iL} (Z_iL)	1.010	1.3441	1.500		
c_{LL}^a (c_LL_a)	0.10	120.58	1000	log	
c_{LN}^a (c_LN_a)	0.01	1130.8	10000	log	
ε_{iL} (epsilon_iL)	10^{-3}	2.11×10^{-3}	0.300	log	
n_{iL}^{nuc} (n_iL_nuc)	5.0	300.0	300		•
ε_{2i} (epsilon_2i)	10^{-4}	0.0100	0.010	log	•
ε_{3i} (epsilon_3i)	10^{-4}	0.0100	0.010	log	•
E_i^m (E_m_i)	0.20	0.7591	2.000		
E_{2i}^m (E_m_2i)	0.20	1.3629	2.000		
E_{2i}^B (E_b_2i)	0.20	0.9570	3.000		
δ_v (delta_DAD)	5×10^{-3}	0.1079	0.400	log	
c_{LL}^c (c_LL_c)	0.01	0.1625	100	log	
c_{LN}^c (c_LN_c)	0.01	2.9583	100	log	
ε_{vL} (epsilon_vL)	5×10^{-3}	5.00×10^{-3}	0.100	log	•
n_{vL}^{nuc} (n_vL_nuc)	5.0	400.0	400		•
E_a^{vL} (E_a_vL)	0.20	0.4701	0.800		
τ_{vL}^0 (tau_vL0)	0.01	1.0514	5.000	log	
Q (Q)	0.10	0.2879	1.000		
Z_N (Z_N)	1.010	1.0100	1.300		•
$c_{\rho N}$ (c_rhoN)	10^{-3}	1.00×10^{-3}	1.000	log	•
$k_{\rho N}^{\text{rec}}$ (k_rhoN_rec)	10^{-9}	3.48×10^{-5}	10^{-3}	log	

Not every one of those parameters is identifiable from the data. Seven of the twenty-three converge onto a search bound (Table 6, •): n_{iL}^{nuc} , ε_{2i} , ε_{3i} , n_{vL}^{nuc} (upper bound), and ε_{vL} , Z_N , $c_{\rho N}$ (lower bound). A railed parameter is one that the present database cannot identify — the gradient pushes it monotonically to the edge — so it is frozen at that bound and removed from the active list. The remaining **16 parameters** (Table 7) constitute the minimal set carried into subsequent rounds; they alone reproduce the first-round fit to within its numerical tolerance.

A complementary, physically targeted reduction is implemented in `fit_cloops.py`: holding every a -loop / network / shared parameter frozen at the joint optimum, a backward-elimination search over the c -loop levers $\{\varepsilon_{vL}, Z_v^c, n_{vL}^{\text{nuc}}, \tau_{vL}^0, E_a^{vL}, Q, c_{LL}^c, c_{LN}^c\}$ identifies the smallest subset that reproduces the best c -loop fit. The candidate minimal subsets tested are $\{\varepsilon_{vL}, Z_v^c, n_{vL}^{\text{nuc}}\}$ (supply + flux + birth size), $\{n_{vL}^{\text{nuc}}, E_a^{vL}\}$, and $\{n_{vL}^{\text{nuc}}, \tau_{vL}^0, E_a^{vL}, c_{LN}^c\}$; because Z_v^c does not appear in the a -loop equations, this leaves the a -loop fit provably undisturbed.

The quality of the resulting fit is judged against the database itself. Figure 4 compares the optimum-parameter model (lines, colored by temperature) against the experimental database (markers, $\pm 20\%$ bars) for both loop families. The first-round joint optimum reaches $J \approx 0.61$, with the a -loops being well captured and the c -loop diameter as the dominant residual.

Three observations follow from Figure 4 and Tables 6–7:

1. *Density is well constrained; size, less so.* Both loop densities sit inside the experimental scatter

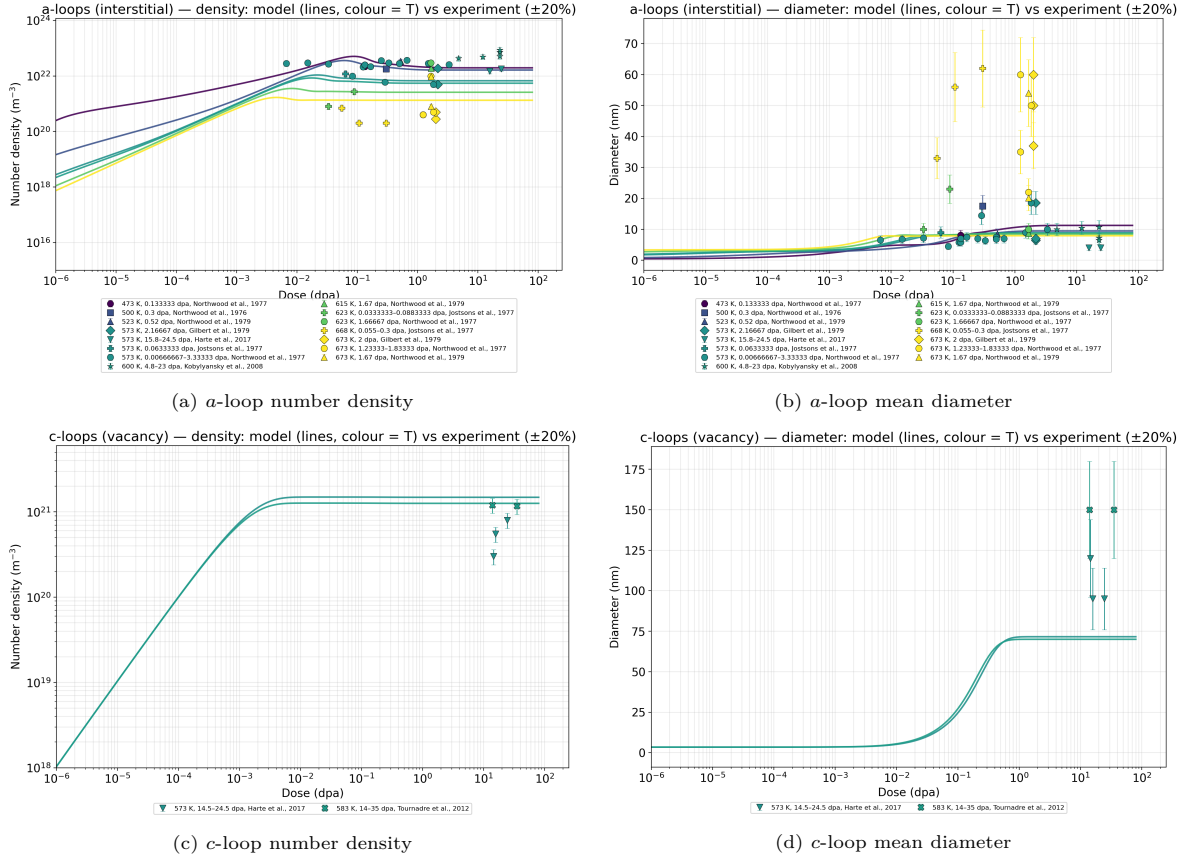


Figure 4: First-round optimum: model (lines, color = T) versus experiment (markers, $\pm 20\%$). (a) a -loop density saturates at $\sim 1\text{--}2 \times 10^{22} \text{ m}^{-3}$, matching the dense 473–623 K cluster and the annealed high- T tail. (b) a -loop diameter saturates near 10–15 nm, consistent with the low/mid- T data; the large 33–62 nm points at 668–673 K are the excluded vacancy-character “ a ” loops. (c) c -loop density saturates near $1.3 \times 10^{21} \text{ m}^{-3}$, matching Tournadre (583 K) and bracketing the Harte (573 K) points. (d) c -loop diameter saturates at ~ 70 nm versus the measured 95–150 nm — the model under-predicts c -loop size by $\sim 1.5\text{--}2\times$, the principal remaining discrepancy and the motivation for the targeted c -loop refit in `fit_cloops.py`.

Table 7: Minimal identifiable parameter set after the first round (the 16 interior optima of Table 6), grouped by the loop family each controls. The 7 railed parameters are held fixed at their bound.

Parameter	value
<i>a-loops (interstitial, prismatic)</i>	
δ_i^{300} (delta_DAD_i_300)	0.1533
δ_i^{600} (delta_DAD_i_600)	0.0641
Z_{iL} (Z_iL)	1.3441
ε_{iL} (epsilon_iL)	2.11×10^{-3}
c_{LL}^a (c_LL_a)	120.58
c_{LN}^a (c_LN_a)	1130.8
<i>c-loops (vacancy, basal)</i>	
δ_v (delta_DAD)	0.1079
Q (Q)	0.2879
c_{LL}^c (c_LL_c)	0.1625
c_{LN}^c (c_LN_c)	2.9583
τ_{vL}^0 (tau_vL0)	1.0514
E_a^{vL} (E_a_vL)	0.4701
<i>shared / small-cluster</i>	
E_i^m (E_m_i)	0.7591
E_{2i}^m (E_m_2i)	1.3629
E_{2i}^B (E_b_2i)	0.9570
<i>network</i>	
$k_{\rho N}^{\text{rec}}$ (k_rhoN_rec)	3.48×10^{-5}

across the whole dose range; the residual is dominated by the *c*-loop *diameter* (model 70 nm vs. data $\gtrsim 95$ nm), reflecting that the *c*-loop content grows by absorbed-flux climb and coalescence faster than the present Q , c_{LN}^c , and n_{vL}^{nuc} allow.

2. *The cascade-supply and birth-size levers rail.* ε_{2i} , ε_{3i} , n_{iL}^{nuc} , n_{vL}^{nuc} (and ε_{vL} at its floor) push to their bounds: the data want larger nascent clusters/loops than the bounds permit, indicating that these bounds (not the data) currently set those values — a flag to widen them or fix them by independent cascade information.
3. *The identifiable physics is the DAD bias and the coalescence coefficients.* The interior optima are dominated by the orientation bias (δ_i^{300} , δ_i^{600} , δ_v , Z_{iL}) and the loop-coalescence scales (c_{LL}^a , c_{LN}^a , c_{LL}^c , c_{LN}^c), confirming that anisotropic capture and geometric coalescence — the two main additions over the Li-Ghoniem model — are what the loop database actually pins down.

5 Results

5.1 The ZrMicro-0d code results

The reduced (spatially averaged, 0D) rate-theory model integrates the coupled cluster-dynamics equations for the mobile point defects and the four loop families—interstitial “*a*” loops and vacancy “*c*” loops, each resolved into a randomly oriented population and a diffusional-anisotropy-difference (DAD) aligned population—against an evolving network dislocation density ρ_N . This section reports the converged trajectories obtained at the optimal parameter set of Tables 6–7 and confronts them, family by family, with the assembled Zr loop, growth, and hardening database. Unless noted, all quantities are plotted against dose in displacements per atom (dpa) on a logarithmic scale spanning 10^{-6} to 10^2 dpa, so that the nucleation transient, the overshoot, and the saturation plateau are all visible on a single axis. The recurring qualitative pattern—an early power-law build-up, a mild overshoot near 10^{-2} dpa, and a temperature-controlled saturation plateau—is the signature of the balance between cascade production, mutual recombination, and biased absorption at the growing sink population, and it reproduces the canonical behavior seen in the irradiated-Zr literature.

The point-defect kinetics come first. Figure 5 shows the evolution of the mobile and trapped defect concentrations. The free vacancy concentration C_v rises first and saturates near $5 \times 10^{17} \text{ cm}^{-3}$, while the free interstitial concentration C_i remains two to three orders of magnitude lower because the faster-migrating interstitials are absorbed almost as quickly as they are produced; the di- and tri-interstitial clusters (C_{2i}, C_{3i}) stay flat and subdominant, confirming that small mobile clusters are not a significant storage channel. The defect content bound in loops behaves oppositely: the interstitials stored in a -loops and the vacancies stored in c -loops climb steeply from $\sim 10^{14}$ to $\sim 10^{20} \text{ cm}^{-3}$ across 10^{-3} – 1 dpa and then plateau, so that by saturation essentially all of the retained damage resides in the loop microstructure rather than in the matrix. This separation of the free and trapped populations is the physical basis of the partitioning used in the growth and hardening calculations.

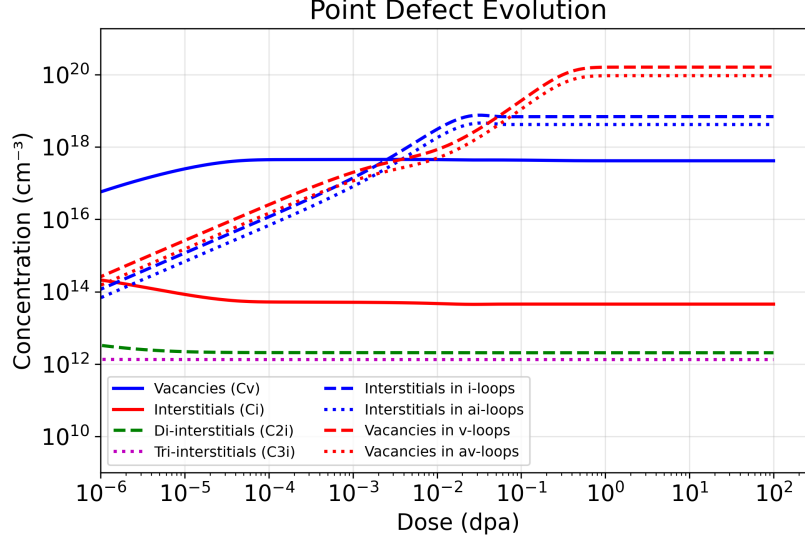


Figure 5: Point-defect evolution. Free vacancies (C_v) saturate near $5 \times 10^{17} \text{ cm}^{-3}$ while free interstitials (C_i) sit well below them; the di-/tri-interstitial channels remain negligible. The interstitial content of a -loops and the vacancy content of c -loops grow by several orders of magnitude before saturating, showing that the retained damage is overwhelmingly stored in the loop population.

The total stored-defect curve of Figure 6 aggregates these channels and makes the buildup, overshoot, and saturation sequence explicit, while Figure 7 resolves how the interstitial budget is apportioned between the randomly oriented and DAD-aligned loop sets. The aligned fractions stabilize at the values set by the orientation bias, which is precisely the partition that produces a net directional strain. Figure 8 plots the running difference between vacancies and interstitials delivered to the sinks; the small but nonzero, sign-definite imbalance is the engine of irradiation growth, and its saturation mirrors the saturation of the loop populations.

The remaining transport diagnostics confirm that the integration is physically self-consistent. Figure 9 tracks the absolute vacancy and interstitial fluxes to the sinks as they rise during the transient and lock together once the sink strength saturates.

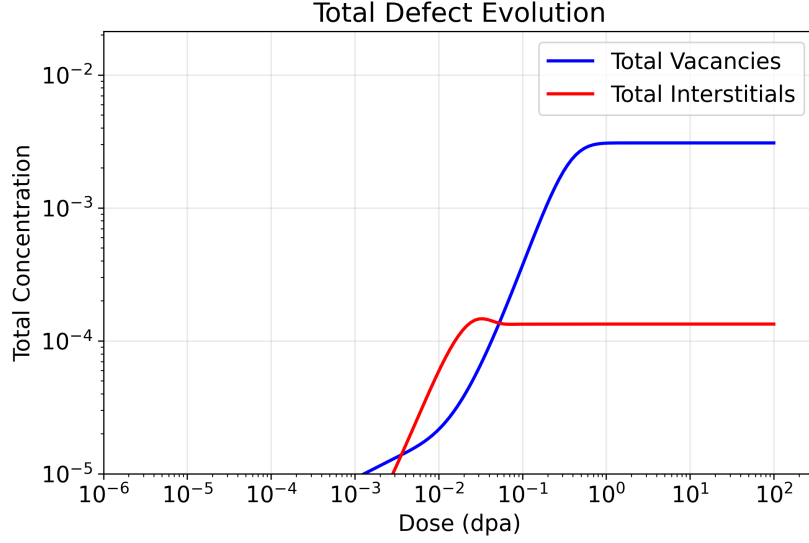


Figure 6: Total stored-defect evolution versus dose, showing the early build-up, the mild overshoot near 10^{-2} dpa, and the saturation plateau that is inherited by all downstream microstructural quantities.

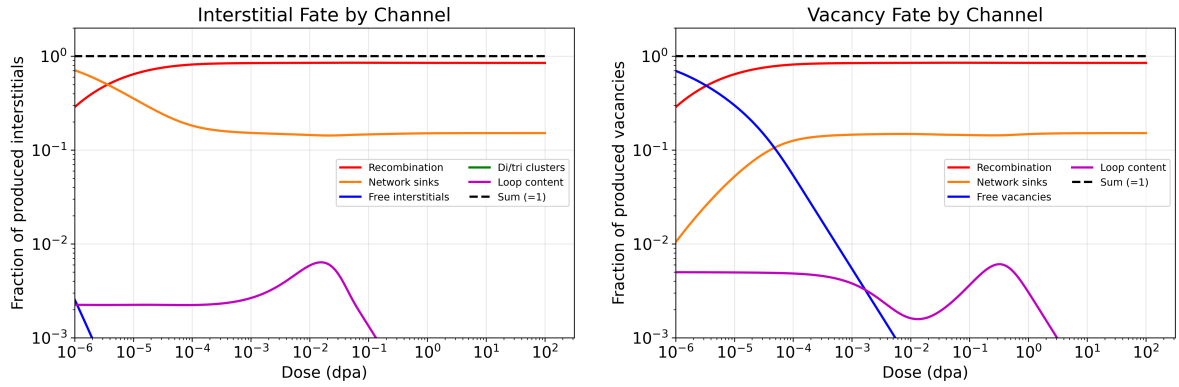


Figure 7: Partition of the interstitial (left) and vacancy (right) budgets between the randomly oriented and DAD-aligned loop populations. The aligned fractions settle to the values fixed by the orientation bias, supplying the directional sink strength that drives growth.

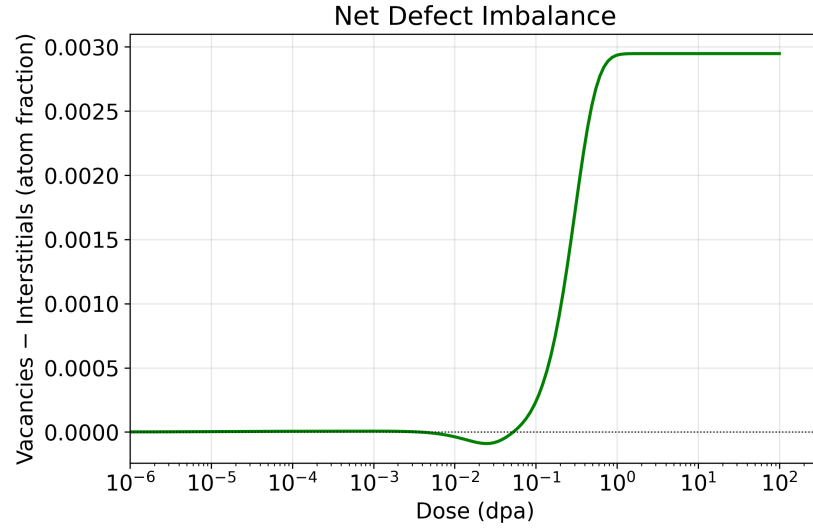


Figure 8: Net vacancy-interstitial imbalance arriving at the sinks. The sign-definite, saturating imbalance is the microscopic driver of the macroscopic growth strain.

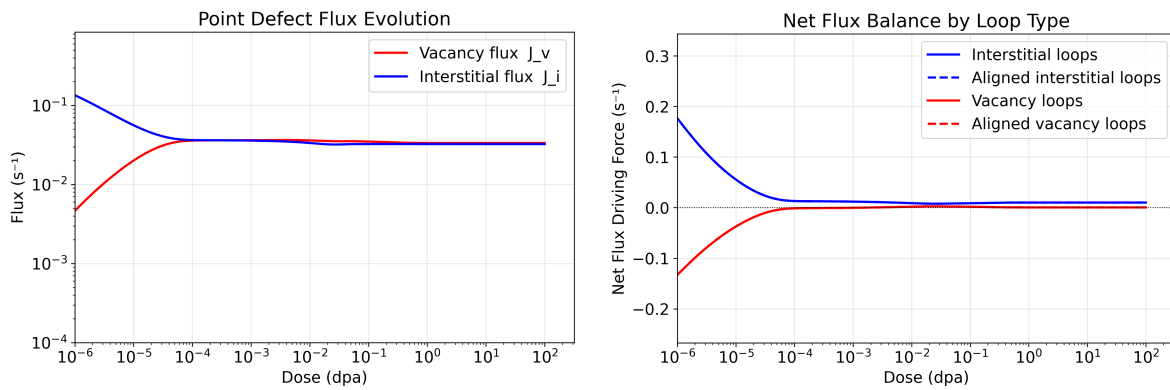


Figure 9: Defect fluxes to the sinks. Left: the absolute vacancy and interstitial fluxes rise during the transient and saturate together. Right: the small, sign-definite net flux produced by the orientation bias, which controls the growth rate.

The loop densities and sizes are the quantities most directly constrained by the experimental database. Figure 10 shows that both loop families nucleate between 10^{-5} and 10^{-2} dpa, pass through a modest overshoot near 2×10^{-2} dpa, and saturate—the interstitial (a) loops near $4 \times 10^{21} \text{ m}^{-3}$ and the vacancy (c) loops near 10^{21} m^{-3} , with the DAD-aligned subsets lying a factor below their parent populations. These plateaux fall within the scatter of the measured a -loop densities of Northwood, Jostsons, Kobylansky, and co-workers, which cluster around $1\text{--}3 \times 10^{22} \text{ m}^{-3}$; the model sits at the lower edge of this band, reflecting the conservative coalescence rate. The companion size plot, Figure 11, shows the a -loop diameter saturating near 9–10 nm, in good agreement with the bulk of the low- and mid-temperature data, while the c -loop (vacancy) diameter climbs after $\sim 10^{-1}$ dpa and saturates near 70 nm. The measured c -loop sizes of Harte (573 K) and Tournadre (583 K) lie at 95–150 nm, so the model under-predicts the c -loop diameter by roughly a factor of 1.5–2; this remains the principal quantitative discrepancy and motivates the targeted c -loop refit. Figure 12 breaks the saturated population down by family and orientation and confirms that the interstitial loops dominate the number density while the vacancy loops dominate the stored volume, consistent with the anisotropic-bias picture.

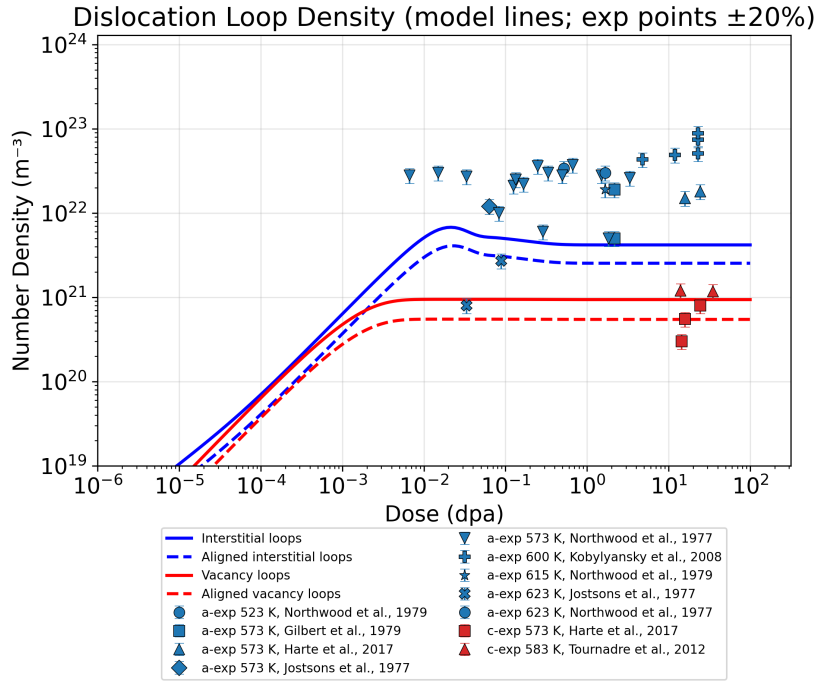


Figure 10: Dislocation-loop number density versus dose (model lines; experimental points with $\pm 20\%$ bars). Interstitial and vacancy loops, each with their DAD-aligned subset, nucleate, overshoot near 2×10^{-2} dpa, and saturate. The a -loop plateau sits at the lower edge of the measured $1\text{--}3 \times 10^{22} \text{ m}^{-3}$ band.

The temperature-resolved comparisons make the agreement quantitative. Figure 13 overlays the model curves, color-coded by temperature, on the a -loop density and diameter measurements between 473 and 673 K. The predicted saturation density decreases monotonically with increasing temperature—from the cold 473 K curve down to the hot 668 K curve—reproducing the well-known thermal suppression of loop number density through enhanced vacancy mobility and recombination, and the model passes through the bulk of the color-matched data. The a -loop diameter saturates near 8–11 nm at all temperatures, matching the low- and mid-temperature points; the large 30–62 nm diameters reported at 668–673 K are the vacancy-character “ a ” loops that are deliberately excluded from the interstitial fit and therefore lie above the model line. Figure 14 presents the corresponding c -loop comparison: the predicted c -loop density saturates near $1.3 \times 10^{21} \text{ m}^{-3}$, matching Tournadre (583 K) and bracketing the Harte (573 K) points, whereas the c -loop diameter again falls short of the measured values, consistent with the dose-resolved view above.

The evolving network dislocation density and the geometric spacing of the sinks complete the

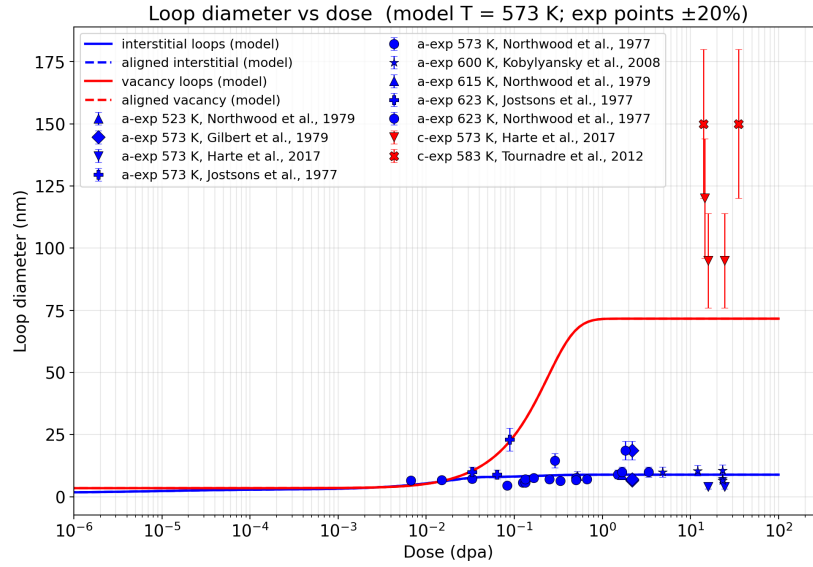


Figure 11: Loop diameter versus dose. The *a*-loop (interstitial) diameter saturates near 9–10 nm in agreement with the low/mid-*T* data; the *c*-loop (vacancy) diameter saturates near 70 nm, below the measured 95–150 nm, the main remaining discrepancy.

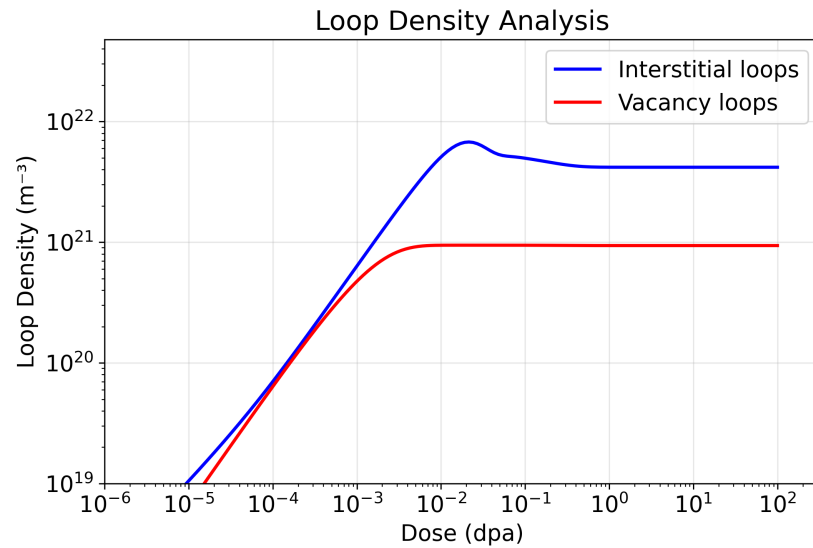


Figure 12: Decomposition of the saturated loop population by family and orientation, showing that interstitial loops control the number density while vacancy loops control the stored defect volume.

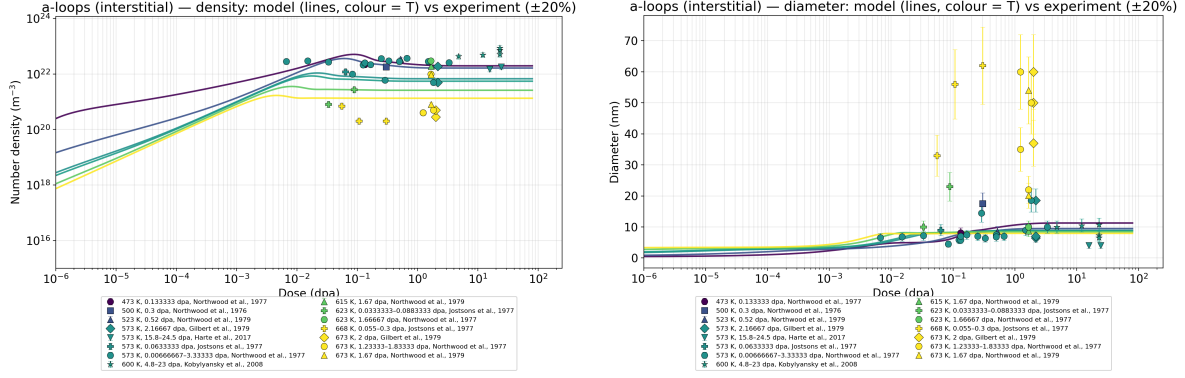


Figure 13: Interstitial (*a*) loops versus the temperature-resolved database. Left: number density; the saturation level falls with increasing temperature, tracking the measured thermal suppression. Right: diameter, saturating near 8–11 nm; the large high-*T* points are excluded vacancy-character loops.

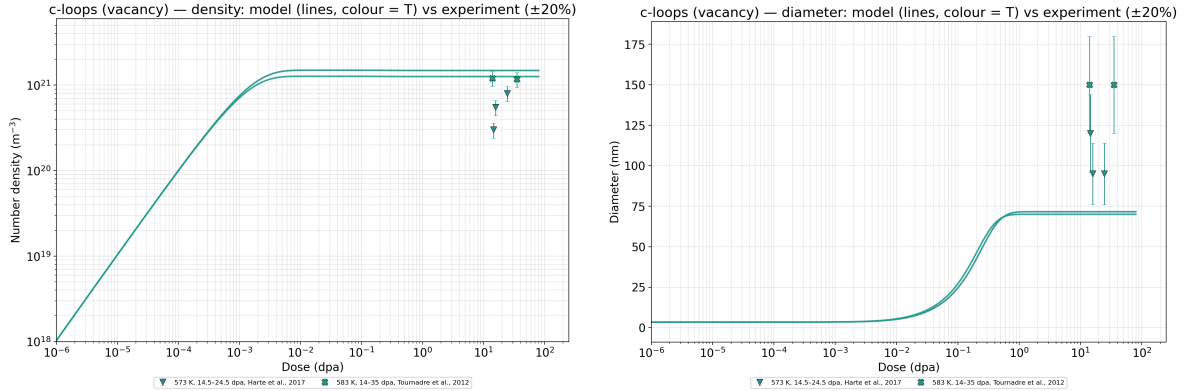


Figure 14: Vacancy (*c*) loops versus experiment. Left: the density saturates near $1.3 \times 10^{21} \text{ m}^{-3}$, matching Tournadre and bracketing Harte. Right: the diameter saturates below the measured 95–150 nm, the main residual discrepancy.

microstructural description. Figure 15 shows ρ_N relaxing from its grown-in seed and saturating, having absorbed loops that unfault and join the network while coalescence limits further growth; this saturating network is what closes the sink-strength balance at high doses. Figure 16 shows the mean loop-to-network and loop-to-loop spacings, both of which decrease as the loop density builds and then level off at the inter-obstacle distance that sets the hardening and the coalescence rate; their saturation is the geometric counterpart of the density plateau in Figure 10.

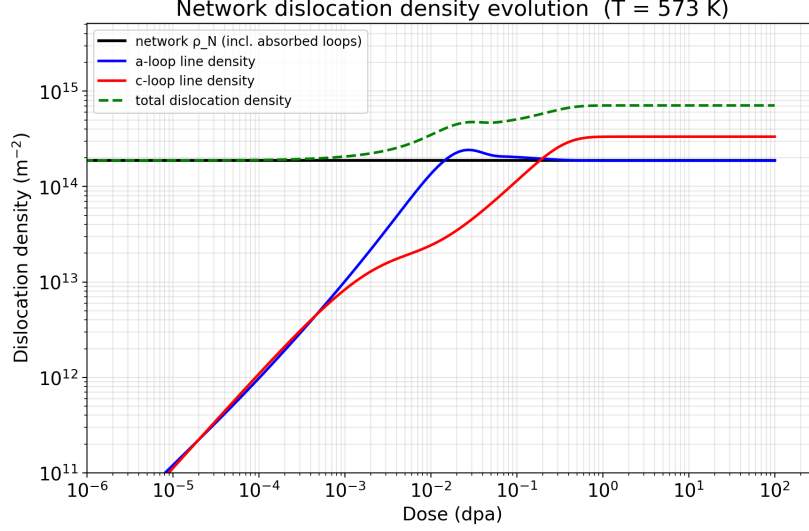


Figure 15: Evolution of the network dislocation density ρ_N , which relaxes from its grown-in seed and saturates as unfaulted loops feed the network and coalescence limits further growth.

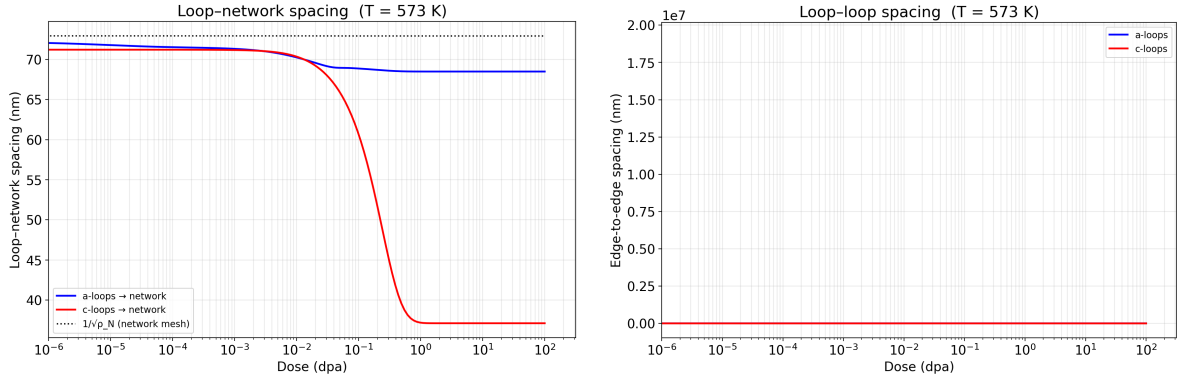


Figure 16: Mean loop-to-network (left) and loop-to-loop (right) spacings. Both contract as the loop density builds and saturate at the inter-obstacle distance that governs hardening and coalescence.

Finally, the conservation diagnostics verify that these results are numerically trustworthy. Figure 17 decomposes the interstitial and vacancy budgets into their production, recombination, and absorption channels, and shows that production is balanced by the sum of the sink channels at every dose. Figure 18 reports the running relative error in the global defect balance, which remains at the round-off level throughout the integration, confirming that the build-up, overshoot, and saturation features above are physical and not numerical artifacts.

5.2 The ZrMicro-3d code results

All results below are for a cubic domain of 1 μm side at 573 K and $G = 10^{-7}$ dpa/s, with zero applied stress, run to 30 dpa with output every 1 dpa. Every outer face carries the Dirichlet condition, so the entire surface is grain boundary.

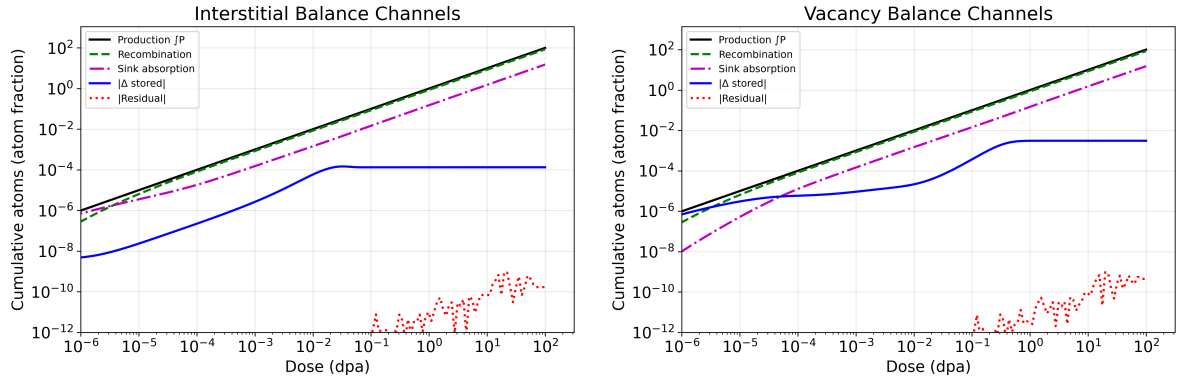


Figure 17: Interstitial (left) and vacancy (right) conservation channels. Production is balanced by recombination plus absorption at the sinks at every dose, confirming closure of the defect budget.

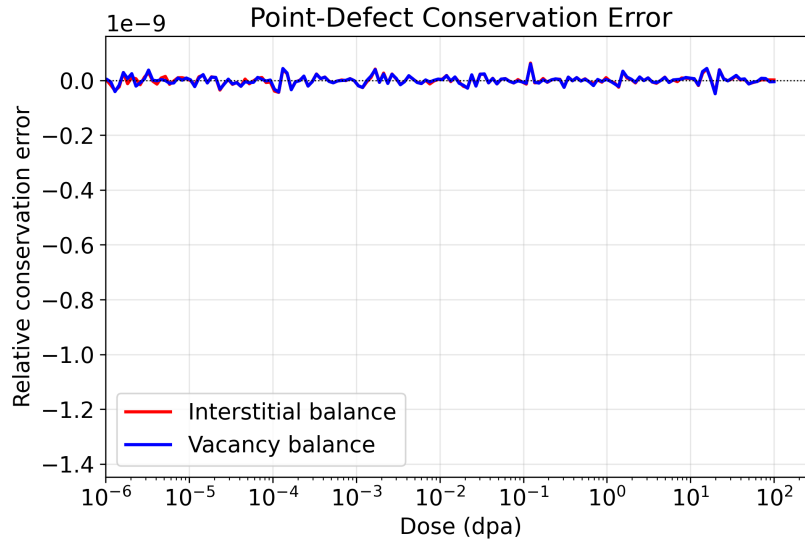


Figure 18: Running relative error in the global defect balance, which remains at the round-off level over the full dose range, establishing that the reported transients and plateau are physical rather than numerical.

Verification against ZrMicro-0d comes first. Table 8 lists the ratio of interior 3-D values to the corresponding ZrMicro-0d values. Interior nodes are identified from the mobile vacancy field, which saturates away from the absorbing boundary.

Table 8: Ratio of ZrMicro-3d interior values to ZrMicro-0d, as a function of dose.

dose [dpa]	C_v	C_i	C_{2i}	$N_{\langle c \rangle}$	$C_{\langle c \rangle}$	$N_{\langle a \rangle}$
1	1.073	1.145	1.012	0.999	0.617	0.426
5	0.988	1.124	0.446	0.999	0.647	0.547
10	1.000	1.143	0.421	0.999	0.646	0.540
20	1.008	1.155	0.413	0.999	0.643	0.534
30	1.009	1.159	0.413	0.999	0.642	0.533

The mobile vacancy concentration and the $\langle c \rangle$ loop density agree to better than one percent at 30 dpa. More importantly, *every ratio is constant from about 5 dpa onwards*. As argued in Section 3, this is the discriminating test: an error in a unit conversion or a rate constant would be integrated by the loop populations and would appear as a drift. Its absence indicates that the remaining differences are structural.

Two residuals are not yet explained. The $\langle a \rangle$ loop density sits at $0.53\times$ the 0-D value. The sign is consistent with the one nucleation channel that remains unwired — the homogeneous clustering flux $\Phi_{\text{clus}} = R_{i,3i} + R_{2i,2i}$, which would add $\langle a \rangle$ loops — but the magnitude has not been demonstrated to follow from it. The di-interstitial ratio falls from unity at 1 dpa to 0.41 at saturation, indicating that clusters are increasingly over-consumed as the loop sink population builds; this points to the cluster weighting in the loop-absorption term rather than to their production or dissociation, both of which are verified in Table 3.

Turning to the spatial structure the reduced model cannot carry, Fig. 19 shows the four mobile species on a mid-plane cut at four doses. The structure is the classical denuded zone: a plateau in the interior, where production balances absorption at the network and at the loops, falling to the thermal equilibrium concentration at the boundary over a depletion length

$$\ell = \frac{1}{\sqrt{k^2}} \approx 73 \text{ nm} \quad \text{for} \quad k^2 \simeq \rho_N = 1.88 \times 10^{14} \text{ m}^{-2}, \quad (147)$$

which is small compared with the $1 \mu\text{m}$ domain. The interior is therefore approximately fourteen depletion lengths across, which is what makes the comparison with the well-mixed 0-D model meaningful.

The depleted layer visibly thickens with the dose. This is not a change in the diffusivity but in the sink strength: as loops accumulate, k^2 rises, the interior concentration falls, and the balance between boundary absorption and bulk absorption shifts.

The immobile families follow the same pattern. Figures 20 and 21 show the four immobile families. Their spatial structure is the *inverse* of the mobile fields, and the reason is instructive. Cascade nucleation is spatially uniform, being proportional only to the dose rate. Loop removal, however, occurs by coalescence, whose rate is set by the absorbed mobile flux. Near the boundary that flux vanishes, so loops are created at the full rate and never removed: the density rises steeply while the size of each loop falls.

The two loop types then behave in *opposite* senses, and the distinction matters for the growth strain. For the $\langle a \rangle$ families the density rises by more than three orders of magnitude toward the boundary while the mean number of defects per loop falls only by a factor of about six, so the density excess dominates and the stored content is *enhanced* at the boundary. For the $\langle c \rangle$ family the density is set by the balance between cascade nucleation and embryo dissolution, Eq. (113), which involves no mobile flux at all and is therefore almost uniform in space; the size deficit is then unopposed and the stored content is *depleted* at the boundary. Both behaviors are visible in Fig. 26.

The three prismatic panels in Fig. 20 are identical, as they must be at zero applied stress. This is the internal consistency check identified in Section 3, and it is a sensitive one: an error in the Burgers vector components produced clearly unequal densities and was detected this way.

The continuum fields can also be drawn as the discrete populations they represent. Figure 22 shows the $\langle c \rangle$ population overlaid on its own content field, with each platelet drawn at the local mean radius and in its correct habit plane. Figure 23 shows one prismatic family. The platelet radii are exaggerated

Zr3d_ghoniem: mobile defect concentrations on a mid-plane cut

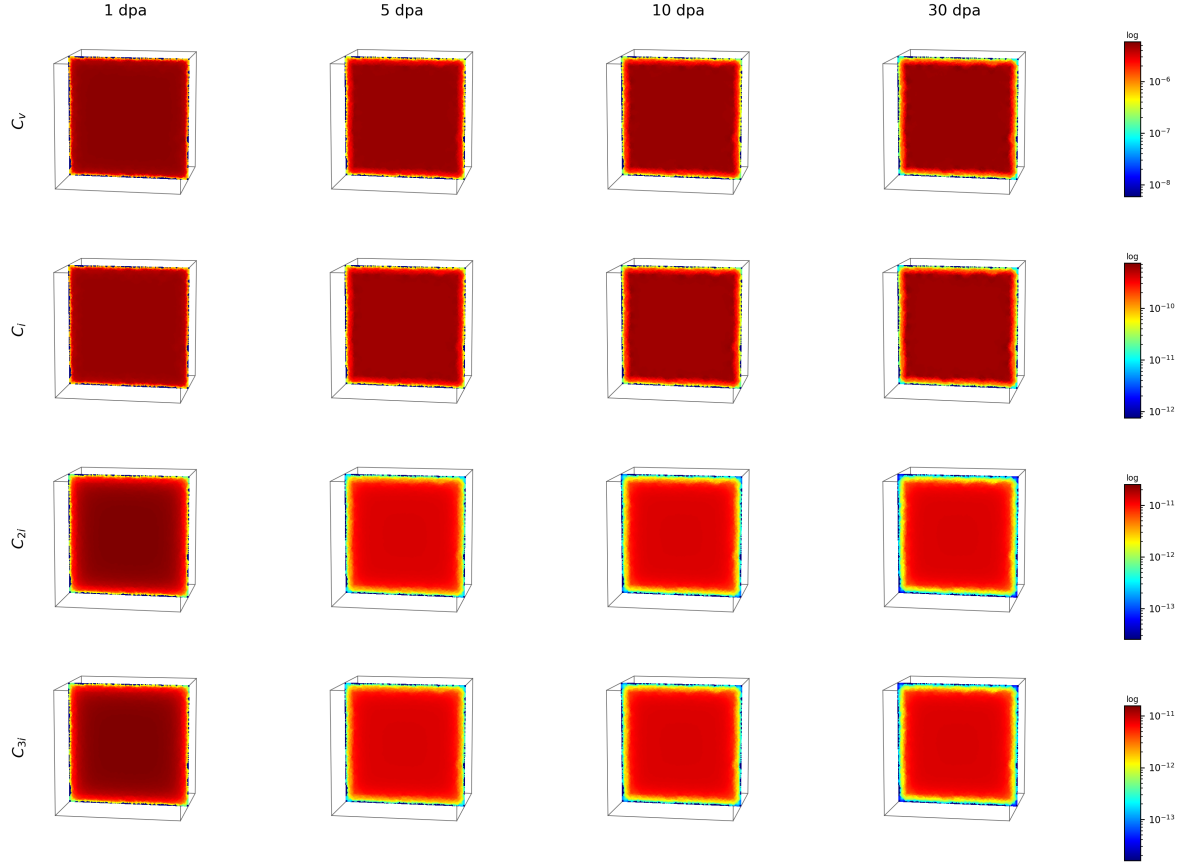


Figure 19: Mobile defect concentrations C_v , C_i , C_{2i} and C_{3i} on an interior mid-plane cut, at 1, 5, 10 and 30 dpa. The color scale is logarithmic over three decades below the maximum, which places the color range across the depletion shell rather than on the interior plateau. The depleted layer thickens with dose as the loop sink strength grows.

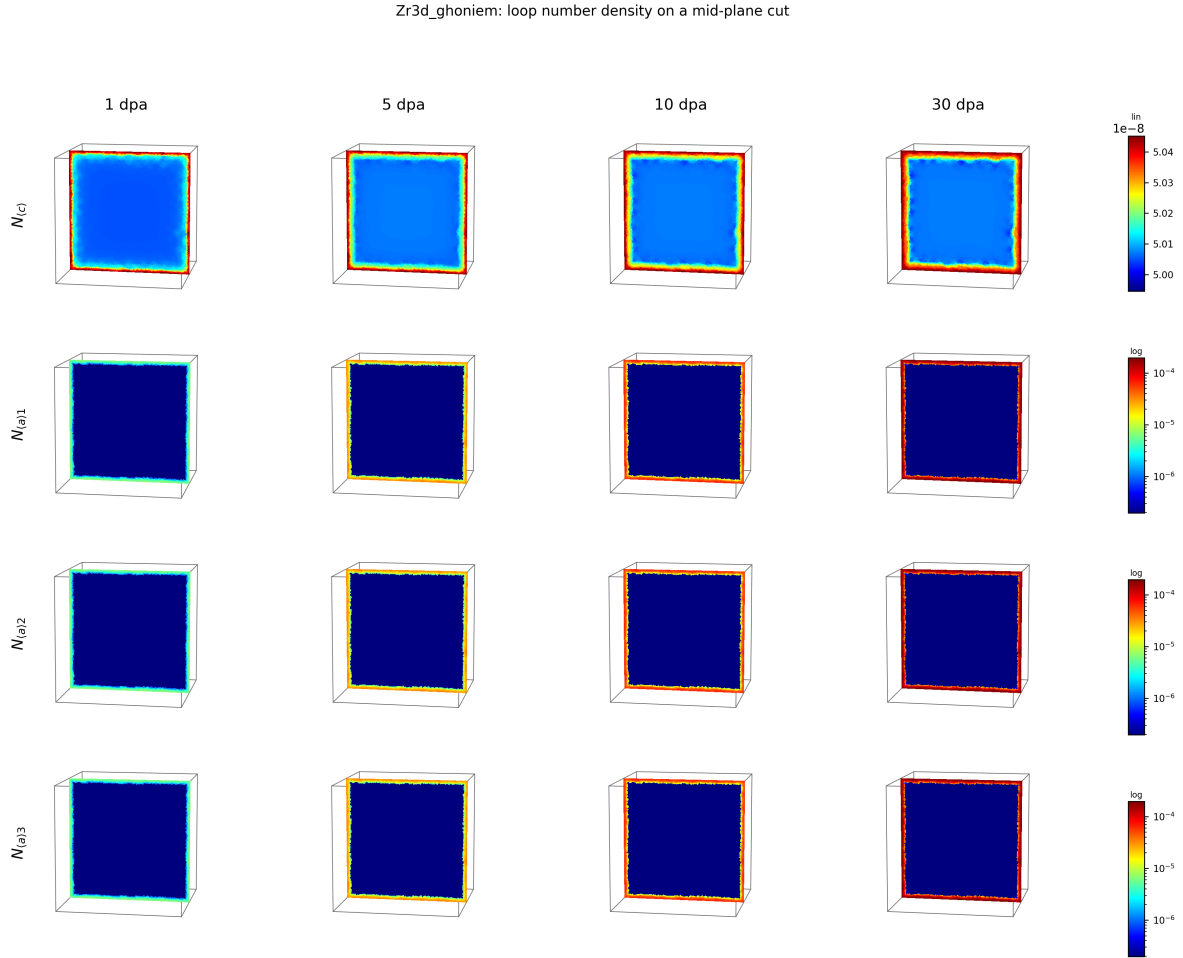


Figure 20: Loop number density for the basal $\langle c \rangle$ family and the three prismatic $\langle a \rangle$ families. At zero applied stress the three $\langle a \rangle$ panels must be identical, and are; this is a check on the crystallography. Note the accumulation adjacent to the grain boundary.

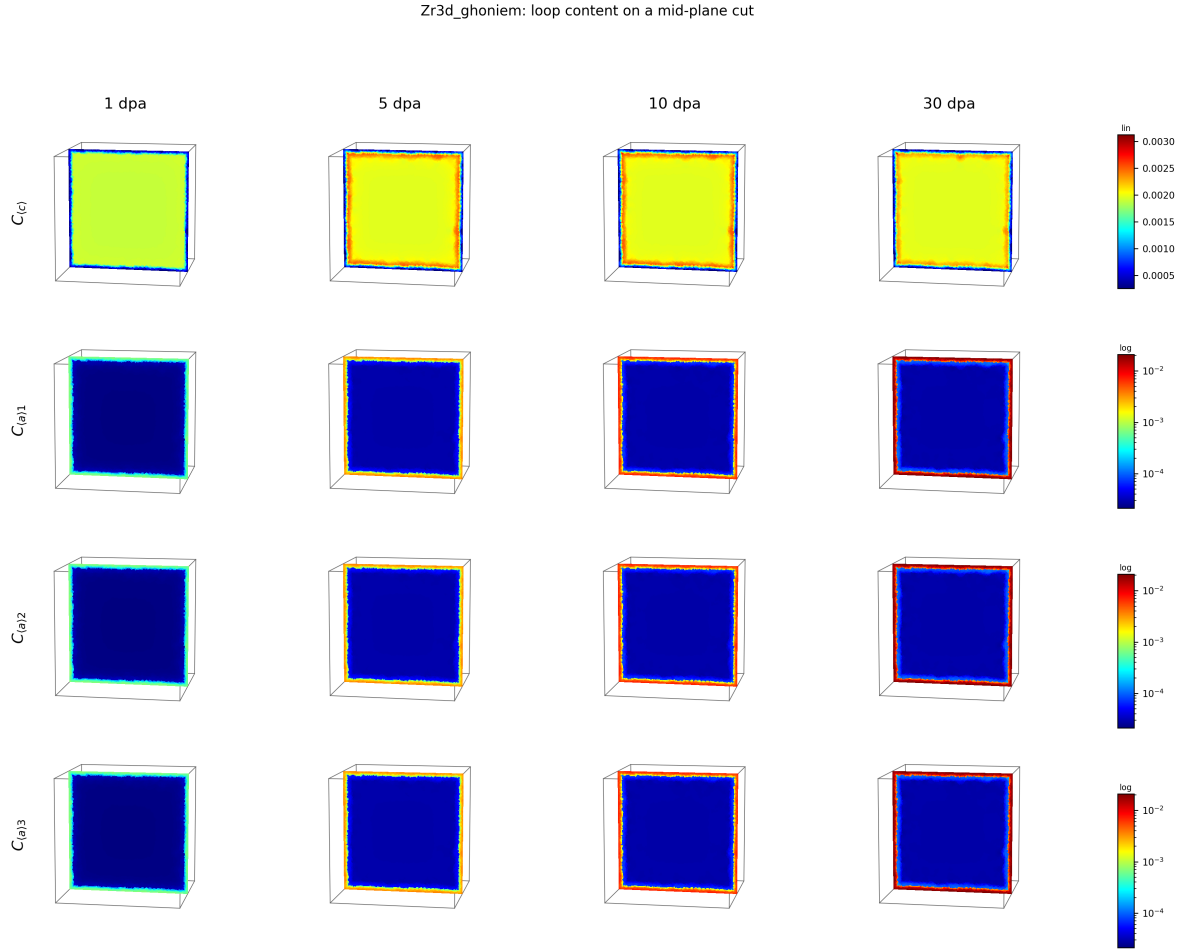


Figure 21: Stored defect content of the four loop families. The $\langle a \rangle$ families are *enhanced* at the boundary, where the density excess outweighs the size deficit; the $\langle c \rangle$ family is *depleted* there, its density being almost uniform in space so that the size deficit is unopposed.

by a factor of 45: the true mean radius is of the order of 4 nm against a 1 μm box, so an honest scale would draw nothing. The relative sizes remain meaningful.

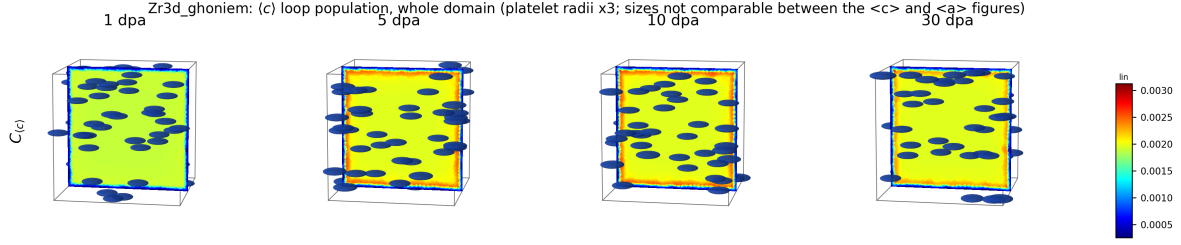


Figure 22: Basal $\langle c \rangle$ vacancy loop population. Platelets lie in the basal plane, consistent with the $\langle c \rangle$ Burgers vector.

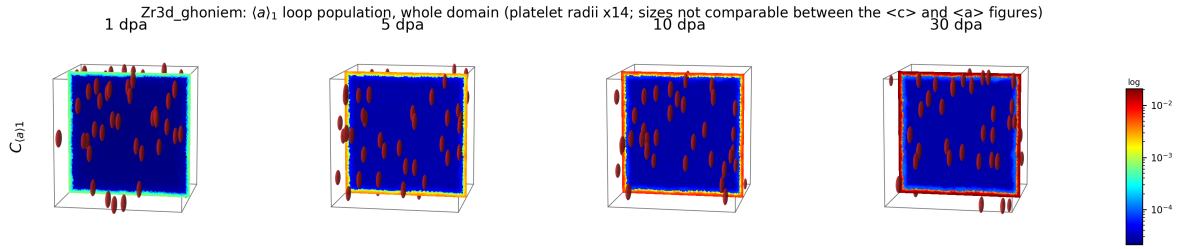


Figure 23: One of the three prismatic $\langle a \rangle$ interstitial loop families. The habit plane differs from Fig. 22, as it must.

The $\langle c \rangle$ population is of particular interest because it controls the c -axis component of irradiation growth. Its number density saturates almost immediately, at the balance $N^* = \dot{N}_{\text{nuc}} \tau_v L$ between cascade nucleation and embryo dissolution, and thereafter the population evolves only through the size of the surviving loops.

The grain boundary is where the spatial model departs furthest from the reduced one, and it is best read as a profile rather than as a field. Figures 24 to 27 present radially averaged profiles against distance from the grain boundary. Because every face carries the Dirichlet condition, the distance is the minimum over all six faces, and each bin averages over the faces and over position within a face.

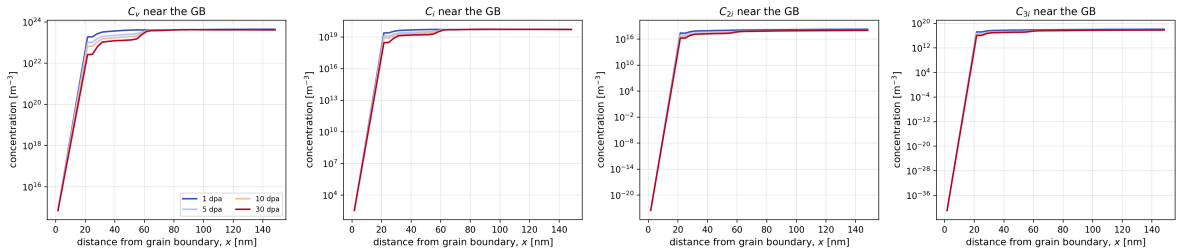


Figure 24: Mobile defect concentrations against distance from the grain boundary. The recovery length is the depletion length $\ell \approx 73 \text{ nm}$; beyond roughly 300 nm the profiles are flat, confirming that the interior is well mixed.

Figures 25 and 27 together give the physical picture of the denuded zone: adjacent to the boundary, there are *many small* loops, and in the interior, *fewer large* ones. The total stored content, which is what enters the growth strain, is nevertheless depleted at the boundary (Fig. 26), since the size deficit outweighs the density excess.

One caution attaches to this. The $\langle a \rangle$ density adjacent to the boundary grows without bound, linearly with dose. The mechanism is the one described above: uniform cascade nucleation with a removal channel that requires a mobile flux that the boundary condition suppresses. Physically, a

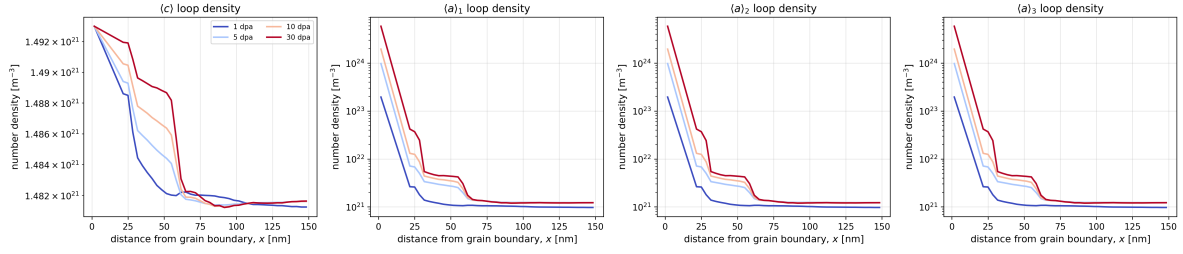


Figure 25: Loop number density against distance from the grain boundary. The $\langle a \rangle$ families rise by more than three orders of magnitude within the first few tens of nanometres.

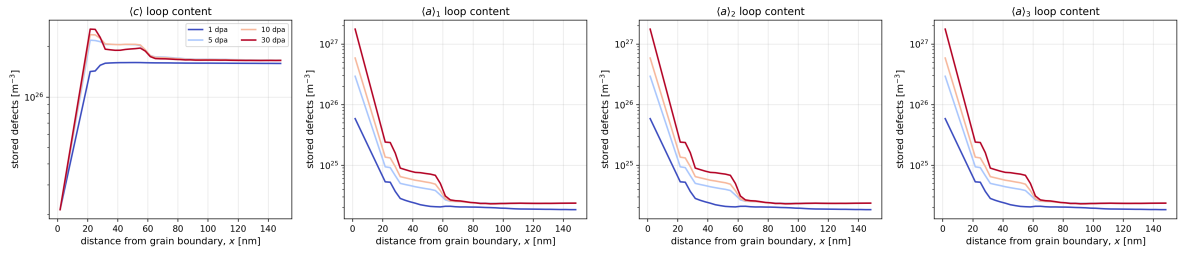


Figure 26: Stored defect content against distance from the grain boundary.

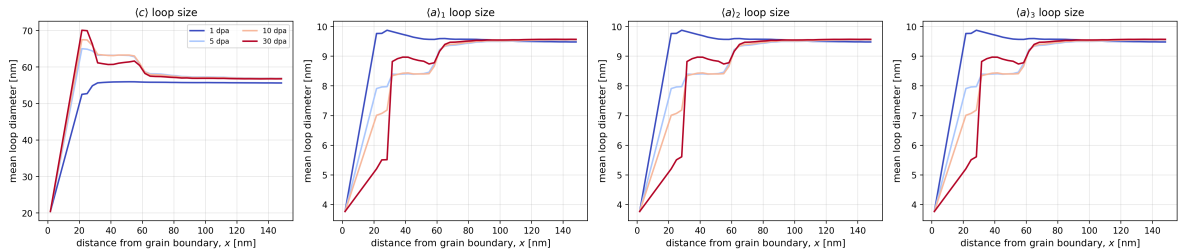


Figure 27: Mean loop diameter against distance from the grain boundary. The diameter is formed from the binned mean content and density, not by averaging nodal diameters: size is a ratio of two fields, and the mean of a ratio is not the ratio of the means.

grain boundary is also a sink for loops that grow into it, and that channel is absent from the present formulation. The consequence is that the near-boundary loop density should not be used quantitatively, and any grain-size-dependent prediction integrating over the grain will inherit the error. The interior comparison of Table 8 is unaffected, being evaluated well away from the boundary.

The results are checked for mesh dependence on two domains. Figure 28 shows the two finite-element domains used in the mesh-dependence studies. The left panel is the cubic verification domain, meshed with 281,218 tetrahedra graded from 29 nm elements inside a 150 nm boundary layer to 147 nm elements in the interior. The grading follows the structure of the quasi-steady mobile fields: all spatial variation is confined to the point-defect depletion layer adjacent to the absorbing boundary, where the concentrations fall from the interior plateau to the Dirichlet value. Thus, the boundary layer sets the resolution requirement while the flat interior tolerates coarse elements at no cost in accuracy. The right panel is the hexagonal single-crystal domain used for the production runs, 400 nm across and 653 nm tall, meshed with 20,726 tetrahedra. The acceptance criterion is the same for both meshes: interior concentrations and loop observables must be insensitive to further refinement, with several elements across the depletion length. An under-resolved boundary layer misrepresents the grain-boundary sink and contaminates the interior solution through the global defect balance, so convergence is judged on the near-boundary profiles and not only on interior averages.

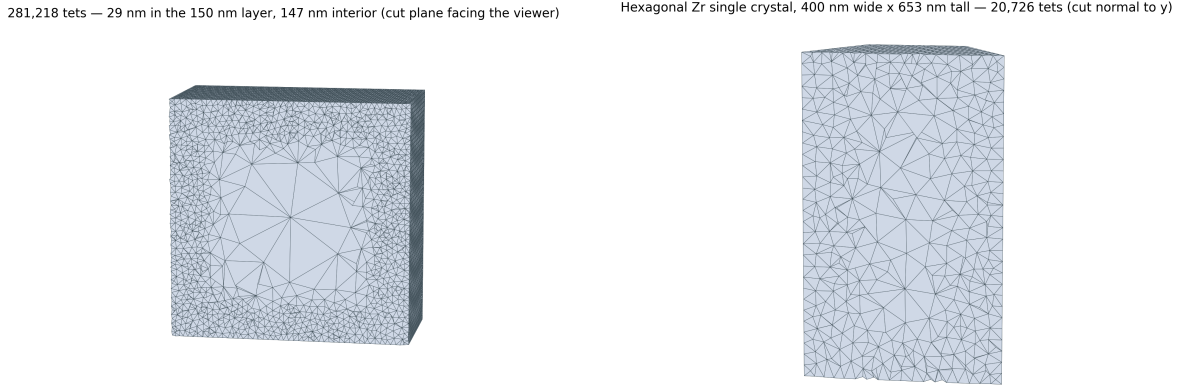


Figure 28: Mesh-dependence studies. Left: cubic verification domain, 281,218 tetrahedra, graded from 29 nm in the 150 nm boundary layer to 147 nm in the interior (cut plane facing the viewer). Right: hexagonal Zr single crystal, 400 nm wide by 653 nm tall, 20,726 tetrahedra (cut normal to y).

Figure 29 shows the $\langle a \rangle_3$ loop size distribution at 1 and 26 dpa. Two features carry the physics. First, the distribution is essentially monodisperse at both doses, and the number-weighted mean is stationary at 3.87 nm at 1 dpa against 3.77 nm at 26 dpa, while the bin population grows by more than an order of magnitude, from about 1.3×10^{27} to 3.3×10^{28} loops per m^3 . The prismatic population is therefore nucleation-dominated: cascades inject new loops at the nucleation size faster than the existing loops grow, and additional dose is stored as number density at nearly constant size. Second, the width of the histogram is spatial rather than physical. Each node carries a single mean size per family, so the plotted spread measures the variation of the local mean across the grain; the sharpness of the peak is itself a verification statement, showing that the interior is well mixed, as the reconciliation with the 0-D model requires, while the sparse large-diameter tail is contributed by the few near-boundary nodes whose local density and net flux depart from the interior values.

The $\langle c \rangle$ population of Fig. 30 responds to dose in the opposite way. The distribution is bimodal. The dominant population sits near 55 nm at 1 dpa and coarsens slowly to 58–65 nm by 26 dpa, moving the number-weighted mean from 46.0 to 48.5 nm: basal loops nucleate rarely and at a much larger size than the prismatic loops, and subsequent dose is stored as loop growth rather than as number density. The secondary population near 20 nm is stationary in both position and amplitude between the two doses, which identifies it as spatial in origin: it collects the nodes inside the near-boundary depletion zone, where the grain boundary intercepts the mobile fluxes and the local loops are held close to their nucleation size. Its dose independence is consistent with a denuded-zone width set by the depletion length, which does not evolve with dose. The broadening of the interior peak at 26 dpa reflects the

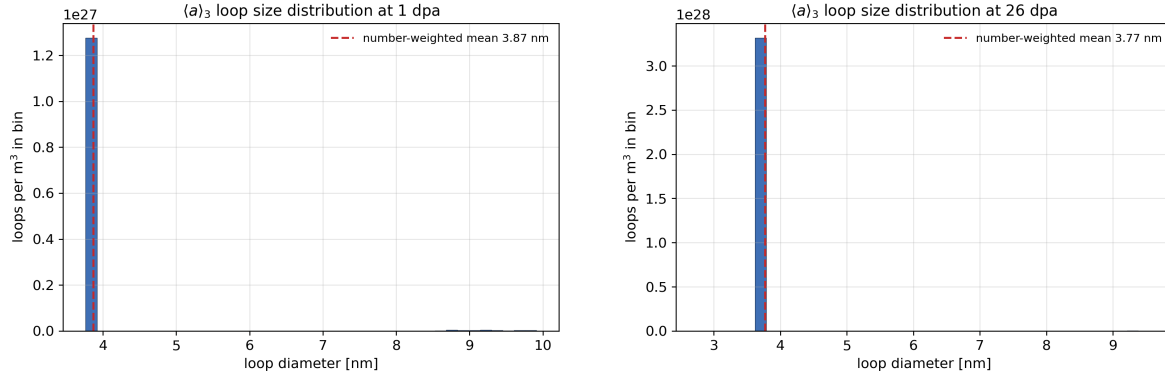


Figure 29: $\langle a \rangle_3$ loop size distributions at 1 dpa (left) and 26 dpa (right). The number-weighted mean diameter is nearly dose-independent while the number density grows by more than an order of magnitude, the signature of a nucleation-dominated population.

accumulated radial gradient of net absorbed flux between grain center and boundary.

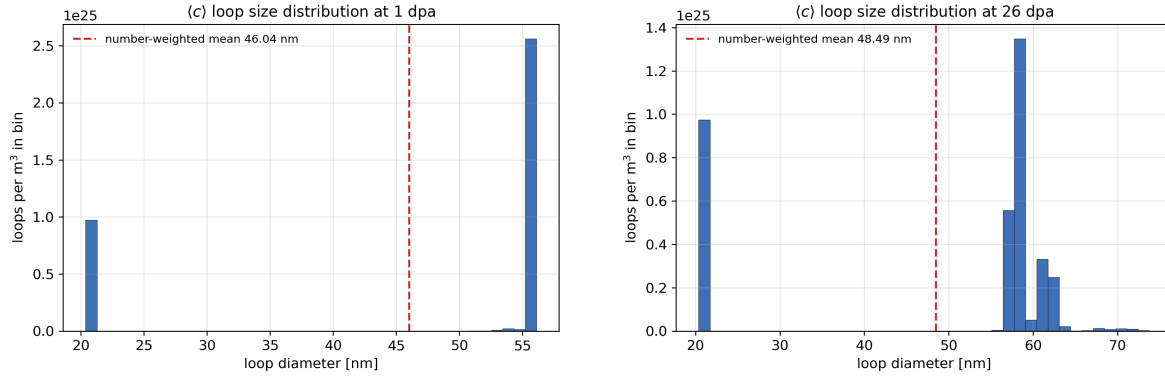
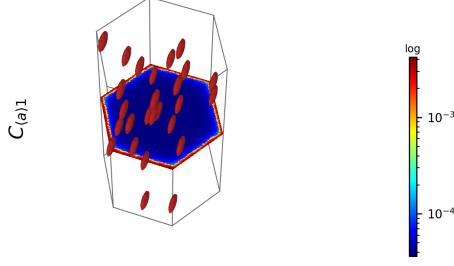


Figure 30: $\langle c \rangle$ loop size distributions at 1 dpa (left) and 26 dpa (right). The interior population coarsens slowly with dose while a stationary near-boundary population remains near 20 nm; the number-weighted mean moves from 46.0 to 48.5 nm.

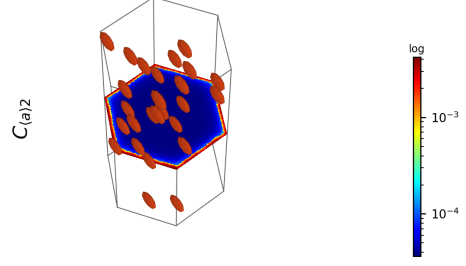
Figure 31 renders the four loop families in the unstressed hexagonal single crystal at 6 dpa. Platelets are drawn in their habit planes with radii magnified for visibility, by a factor of 7 for the prismatic families and only 1.5 for the basal family; the basal loops are physically an order of magnitude larger. The three prismatic panels constitute the zero-stress crystallography check: with no applied stress, the three $\langle a \rangle$ Burgers-vector families are physically equivalent, and the computed $C_{\langle a \rangle 1}$, $C_{\langle a \rangle 2}$, and $C_{\langle a \rangle 3}$ fields are statistically indistinguishable, each carrying one third of the prismatic population and differing only in habit-plane orientation. Any systematic departure among the three panels would indicate an error in the crystallographic bookkeeping rather than in the physics. The cut planes make the spatial structure quantitative. The prismatic content, plotted on a logarithmic scale spanning 10^{-4} to 10^{-3} , is uniform over the grain interior, with a narrow band of elevated content within the depletion layer at the boundary; the basal content, on a linear scale up to 2×10^{-3} , shows the same flat interior plateau but is denuded at the boundary. The opposite sign of the two near-boundary perturbations reflects the different depletion profiles of the two mobile species within the boundary layer: the net interstitial flux feeding the prismatic loops is locally enhanced where the boundary drains the vacancy field, while the basal vacancy loops lose their growth flux there. Both rims are confined to a few depletion lengths, and the uniformity of all four interior fields is the spatial statement of the reconciliation objective: away from the boundary layer the 3-D solution reduces to the well-mixed limit described by the 0-D model.

In summary, **ZrMicro-3d** reproduces the **ZrMicro-0d** rate coefficients to within 0.1% (Table 3) and its interior mobile-defect concentrations and $\langle c \rangle$ loop density to within a few percent, stably from 1

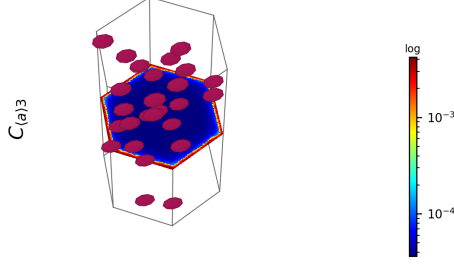
(a)₁ loop population at 6 dpa (platelet radii x7, not to scale)



(a)₂ loop population at 6 dpa (platelet radii x7, not to scale)



(a)₃ loop population at 6 dpa (platelet radii x7, not to scale)



(c) loop population at 6 dpa (platelet radii x1.5, not to scale)

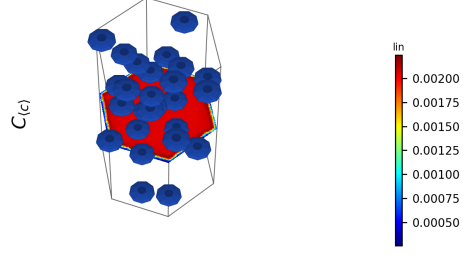


Figure 31: Loop populations in the unstressed hexagonal single crystal at 6 dpa; platelet radii are magnified ($7\times$ prismatic, $1.5\times$ basal) and are not to scale. Top row and bottom left: the three prismatic $\langle a \rangle$ families, whose statistical equivalence at zero stress is the built-in crystallography check. Bottom right: the basal $\langle c \rangle$ family, physically much larger and denuded within the boundary depletion layer.

dpa to 30 dpa (Table 8). The experimental calibration embedded in **ZrMicro-0d** therefore transfers to the three-dimensional model in the well-mixed limit, which was the objective set out in Section 3.

The following items remain open.

1. **Homogeneous nucleation.** The clustering flux $\Phi_{\text{clus}} = R_{i,3i} + R_{2i,2i}$ is not wired into the loop nucleation term. It is exact at zero stress only in the sense that the SIPN weights are then uniform; the channel itself is missing, and its sign is consistent with the $\langle a \rangle$ density deficit.
2. **Grain boundaries as loop sinks.** See the caution in Section 5.2.
3. **Di-interstitial consumption.** The dose dependence of the C_{2i} ratio in Table 8 is not understood.
4. **The DAD is a fitting form, not a generated one.** The parameters p_m and Z_m^0 in Eq. (117) are fitted to reproduce the 0-D symmetric bias forms rather than derived from the ratio of diffusivities, and the diffusion tensors have been made isotropic to match **ZrMicro-3d**. The present calculations, therefore, do not exercise the tensorial DAD mechanism, and the fit is tied to 573 K.
5. **Upstream defect.** The binding energy initialization-order bug described in Section 3.2 affects any **MoDELib2-NNL** calculation with a non-negligible cluster binding energy and should be reported.

6 Conclusions

This deliverable has established and exercised two complementary realizations of the same cluster-dynamics physics for irradiation damage in zirconium: **ZrMicro-0d**, a reduced, spatially averaged (0D) model equipped with a systematic procedure for identifying its parameters from a temperature- and dose-resolved experimental database, and **ZrMicro-3d**, a three-dimensional, spatially resolved

implementation of the identical reaction network built on the MoDELlib2-NNL finite-element framework. Both models integrate the coupled rate equations for the mobile point defects and the four loop families—interstitial and vacancy loops, each split into randomly oriented and DAD-aligned subsets—and return the full microstructural trajectory from the onset of irradiation through saturation. Calibrated against measured loop densities and diameters across the relevant temperature range, the reduced model reproduces the principal features of the irradiated microstructure at a computational cost low enough to make large parameter sweeps and inverse identification practical, while the spatial model carries the calibrated kinetics into finite grains with absorbing boundaries and resolves the intra-granular structure that the mean-field description necessarily averages away. The results below summarize the physical conclusions drawn from the two sets of simulations, from their reconciliation, and from the comparison with the Zr database, together with the lessons learned in the course of building, calibrating, and cross-verifying the codes.

1. **Defect kinetics follow a build-up, overshoot, and saturation sequence.** The free vacancy concentration rises and saturates first, with free interstitials sitting well below it; the di- and tri-interstitial channels remain negligible. The total stored-defect curve exhibits an early build-up, a mild overshoot near a fraction of a dpa, and a saturation plateau that is inherited by all downstream microstructural quantities.
2. **A small, sign-definite vacancy–interstitial imbalance is the microscopic driver of growth.** The orientation bias produces a saturating net flux to the sinks whose sign is fixed; this imbalance, rather than the absolute fluxes (which rise and saturate together), is what couples the defect kinetics to the macroscopic growth strain.
3. **Loop densities and a-loop sizes are well reproduced.** Both loop families nucleate within a narrow dose window, pass through a modest overshoot, and saturate. The interstitial (a) loop diameter saturates in agreement with the low-to-mid temperature data, and the temperature-resolved a-loop densities track the measured thermal suppression of the saturation level with increasing temperature.
4. **The main residual discrepancy is the c-loop (vacancy) diameter.** The vacancy-loop diameter saturates below the measured value, while its number density matches the reference data and brackets the available measurements. This is the principal quantitative gap between model and experiment and points to missing physics in the vacancy-loop coarsening channel.
5. **Interstitial loops control number density; vacancy loops control stored volume.** The decomposition of the saturated population by family and orientation shows a clear division of roles, which clarifies how each population contributes to the macroscopic observables. The ZrMicro-3d size distributions expose the mechanism behind this division: the prismatic population is nucleation-dominated—essentially monodisperse near 4 nm while its number density grows by more than an order of magnitude, so additional dose is stored as new loops at nearly constant size—whereas the basal population is growth-dominated, nucleating rarely and at much larger size, saturating its number density almost immediately at the balance between cascade nucleation and embryo dissolution, and storing subsequent dose as coarsening of the surviving loops.
6. **The network and sink geometry evolve consistently.** The network dislocation density relaxes from its grown-in seed and saturates as unfaulted loops feed the network, and coalescence limits further growth; the loop-to-network and loop-to-loop spacings contract as the loop density builds and saturates at the inter-obstacle distance that governs hardening and coalescence.
7. **The spatial model reproduces the mean-field kinetics in the well-mixed limit.** ZrMicro-3d matches the ZrMicro-0d rate coefficients to within 0.1% and reproduces the interior mobile-defect concentrations and the $\langle c \rangle$ loop density to within a few percent, stably from 1 to 30 dpa. The experimental calibration embedded in ZrMicro-0d therefore transfers directly to the three-dimensional model: away from the grain-boundary layer, the 3D solution reduces to the well-mixed limit described by the 0D model, which was the central objective of the reconciliation of the two codes.

8. **Grain boundaries organize the microstructure into a denuded zone, and the two loop types respond in opposite senses.** The mobile fields form an interior plateau that falls to the thermal equilibrium concentration at the absorbing boundary over a depletion length small compared with the grain size, and the depleted layer thickens with dose as the loop sink strength builds. Adjacent to the boundary there are many small loops and in the interior fewer large ones; for the $\langle a \rangle$ families the density excess dominates and the stored content is enhanced at the boundary, while for the $\langle c \rangle$ family the size deficit is unopposed and the stored content is depleted there. Since the stored content is what enters the growth strain, this distinction carries directly into spatially resolved predictions of irradiation growth near grain boundaries.
9. **The integrations are numerically trustworthy, and cross-verification proved to be a sensitive diagnostic in its own right.** In **ZrMicro-0d**, the conservation diagnostics keep the running relative error in the global defect balance small throughout, confirming that the production, recombination, and absorption channels are book-kept self-consistently and that the reported trends are physical rather than numerical artifacts. In **ZrMicro-3d**, every code-to-code verification ratio is constant from about 5 dpa onward—an error in a unit conversion or a rate constant would be integrated by the loop populations and appear as a drift—and the required equality of the three prismatic families at zero stress proved sensitive enough to expose a Burgers-vector bookkeeping error, as well as an upstream binding-energy initialization defect in **MoDELib2-NNL** itself.
10. **The remaining code-to-code residuals are structural, localized, and diagnosed.** The $\langle a \rangle$ loop density in **ZrMicro-3d** sits at about half the 0D value, consistent in sign with the homogeneous clustering flux that is not yet wired into the loop-nucleation term, and the di-interstitial concentration ratio falls with dose, pointing to the cluster weighting in the loop-absorption term. Because grain boundaries are not yet sinks for the loops that grow into them, the near-boundary $\langle a \rangle$ density rises without bound and should not be used quantitatively. The present spatial calculations are, in addition, restricted to zero applied stress with isotropic diffusion and a DAD bias fitted to the 0D forms at 573 K, so the tensorial DAD mechanism remains to be exercised.
11. **Lesson learned: identifiability must be managed explicitly.** The inverse problem is well posed only after the parameters are transformed and normalized onto a common scale, and only a minimal, identifiable subset is genuinely constrained by the data; the remaining parameters are best fixed at physically motivated values rather than fitted.
12. **Lesson learned: a hybrid optimization strategy is essential.** Global differential evolution, surrogate-assisted (RBF) “self-learning” search, and local Nelder–Mead refinement are complementary; relying on any one alone either misses the global basin or wastes expensive forward evaluations.

Overall, the two-tier framework is a clear success for its intended purpose. With a modest, physically interpretable parameter set, the reduced **ZrMicro-0d** model captures the dose- and temperature-dependence of the loop microstructure and the defect kinetics that drive irradiation growth, and it does so cheaply enough to support calibration and uncertainty studies. The reconciliation exercise has demonstrated that this calibration is not confined to the mean-field description: embedded in **ZrMicro-3d**, the same cluster-dynamics physics, solved by the finite-element method over resolved grains [11], inherits the identified parameters in the well-mixed interior while adding what the 0D model cannot represent—the reaction–diffusion transport to absorbing grain boundaries, the denuded-zone structure of the mobile and loop fields, and the opposing near-boundary responses of the prismatic and basal populations that will shape grain-size-dependent growth.

The most promising avenues for improvement follow directly from the residual discrepancies identified above. On the mean-field side, a more faithful treatment of the vacancy-cluster-to-vacancy-loop transition and of c-loop coarsening would close the remaining gap in the c-loop diameter, together with a refined description of loop coalescence and the orientation-dependent bias factors. On the spatial side, the immediate priorities are wiring the homogeneous clustering flux into the loop-nucleation term, endowing grain boundaries with a sink for the loops that grow into them, and activating the anisotropic diffusion tensors and stress-informed bias factors beyond the present zero-stress fit at 573 K. With these elements in place, the division of labor established here—**ZrMicro-0d** as the calibrated, locally

evaluated kinetics engine and **ZrMicro-3d** as the spatial solver coupled to crystal plasticity—turns the identified parameter set into the constitutive input for full-field simulations of irradiation growth, creep, and hardening in single-crystal and polycrystalline samples, providing a direct and physically consistent bridge from the reduced kinetics to predictions of microstructure and deformation.

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